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Kato

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(54) **PLASMA PROCESSING APPARATUS**

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Oct. 11, 2022 (JP) 2022-163154

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(52) **U.S. Cl.**
CPC ... **H01J 37/32642** (2013.01); **H01J 2237/024**
(2013.01); **H01J 2237/334** (2013.01)

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2237/334; **H01J 2237/332-3348**
USPC **156/345.3**
See application file for complete search history.

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(57) **ABSTRACT**

There is a plasma processing apparatus, comprising: a plasma processing chamber; a substrate support having a ring supporting surface; an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the through holes having upper and lower hole portions, the lower hole portion having a flaring shape; a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the grooves corresponding to the through holes; at least three lift pins disposed below the ring supporting surface, the lift pins corresponding to the grooves, each of the lift pins having upper and lower supporting portions, the upper supporting portion being configured to support the conductive ring, the lower supporting portion configured to support the insulating ring; and at least one actuator configured to vertically move the pins.

19 Claims, 23 Drawing Sheets

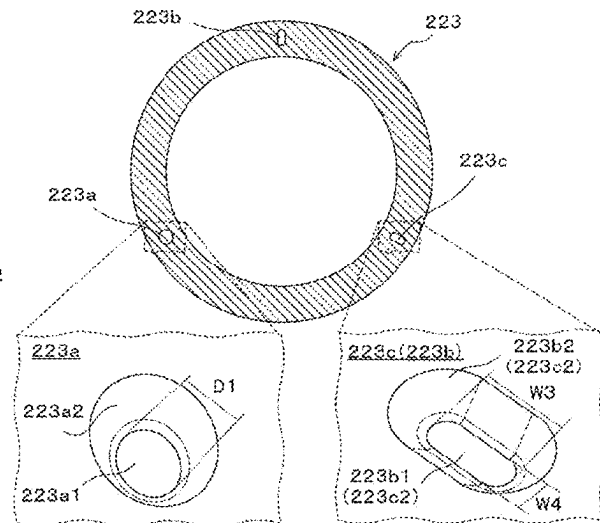
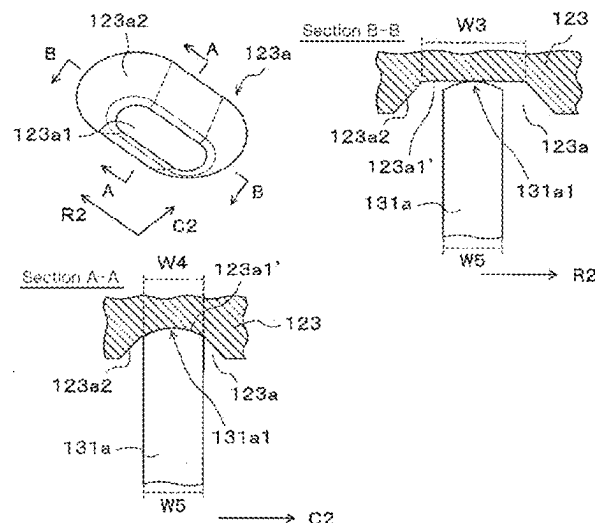


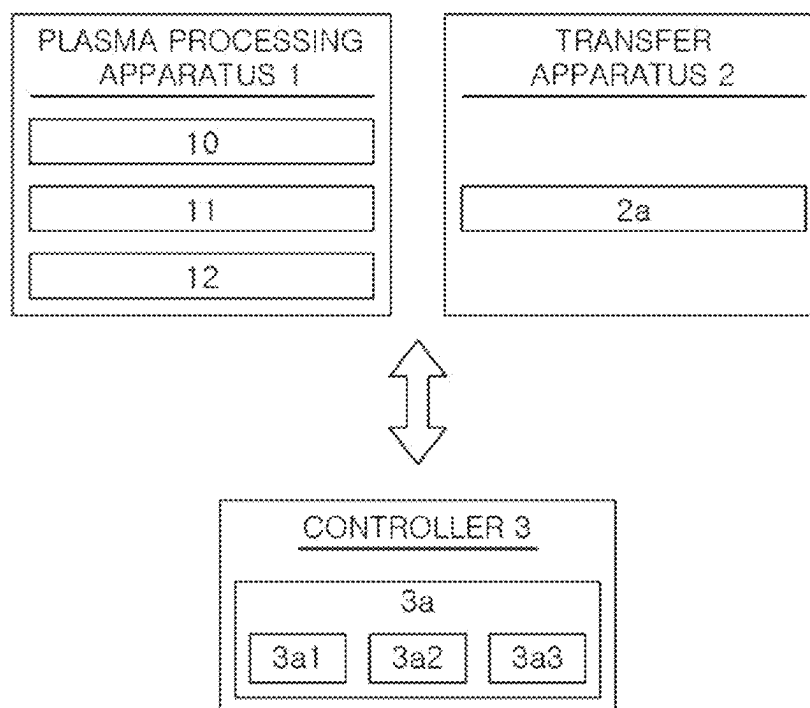
FIG. 1

FIG. 2

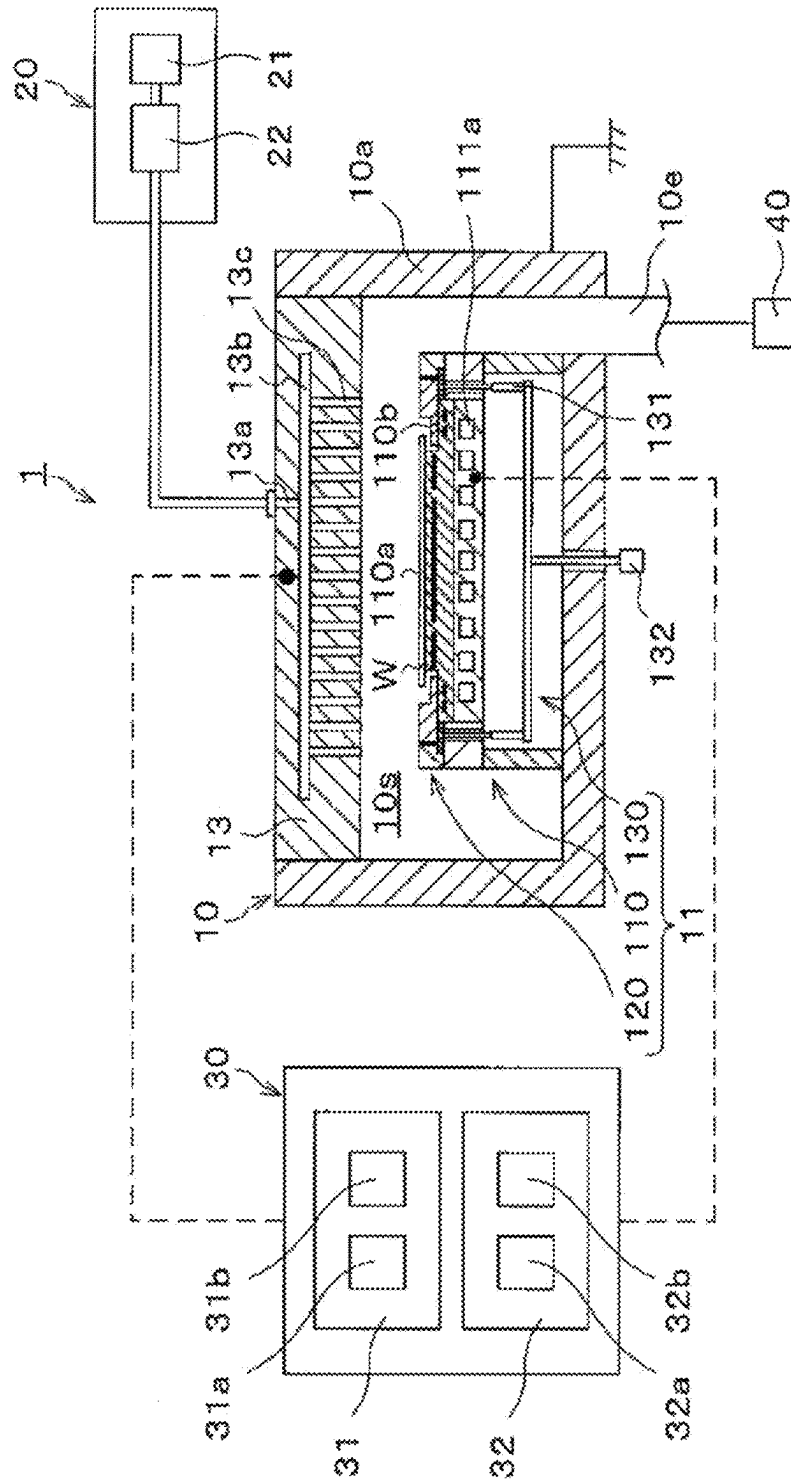


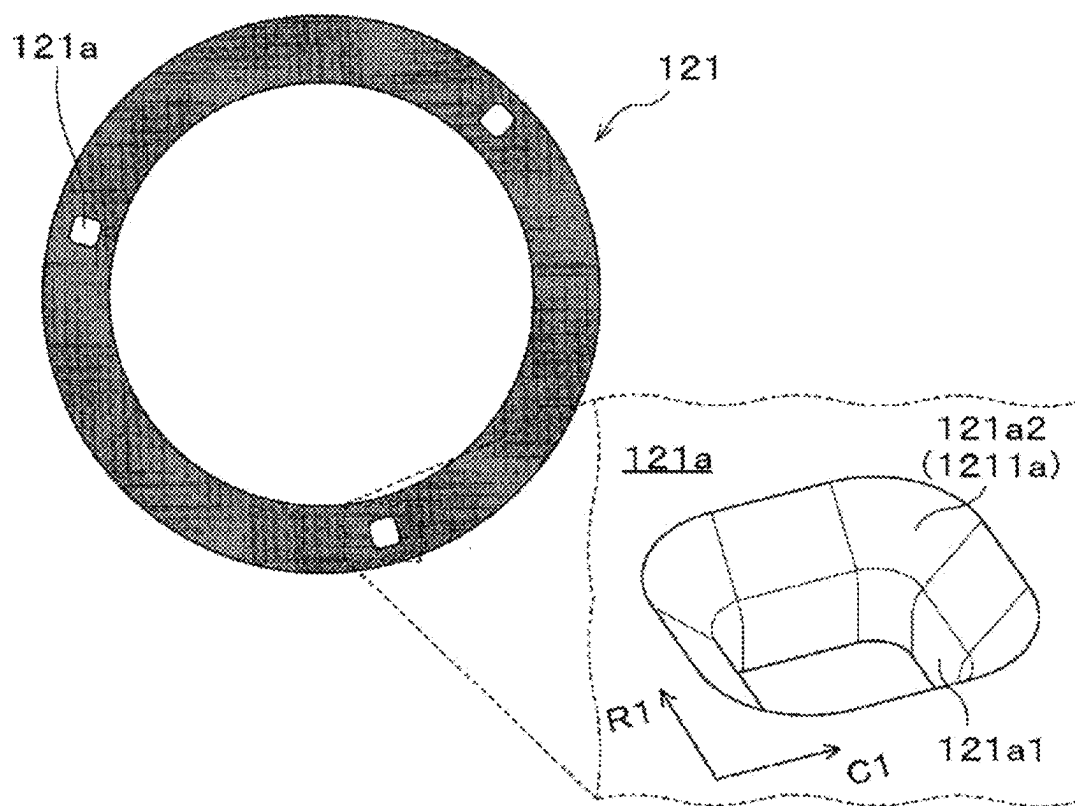
FIG. 4A

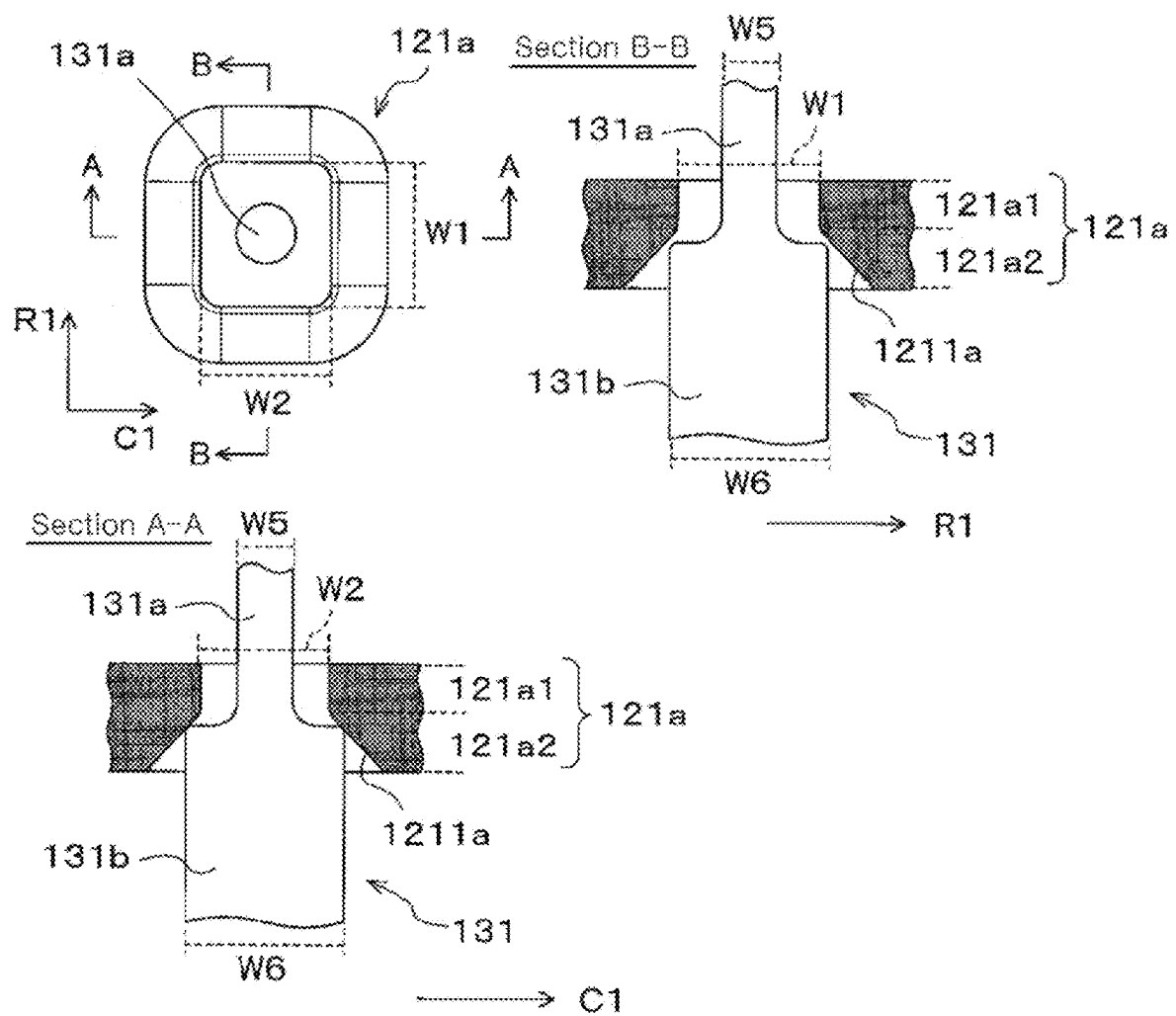
FIG. 4B

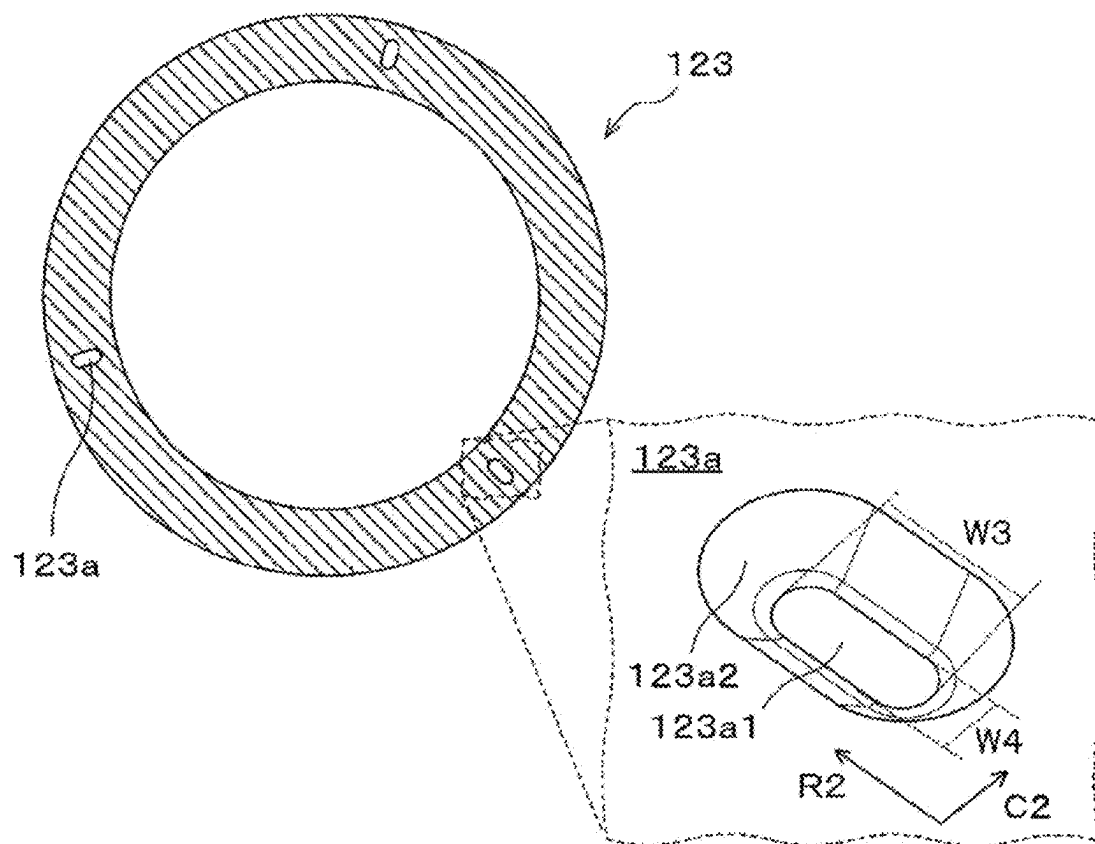
FIG. 5A

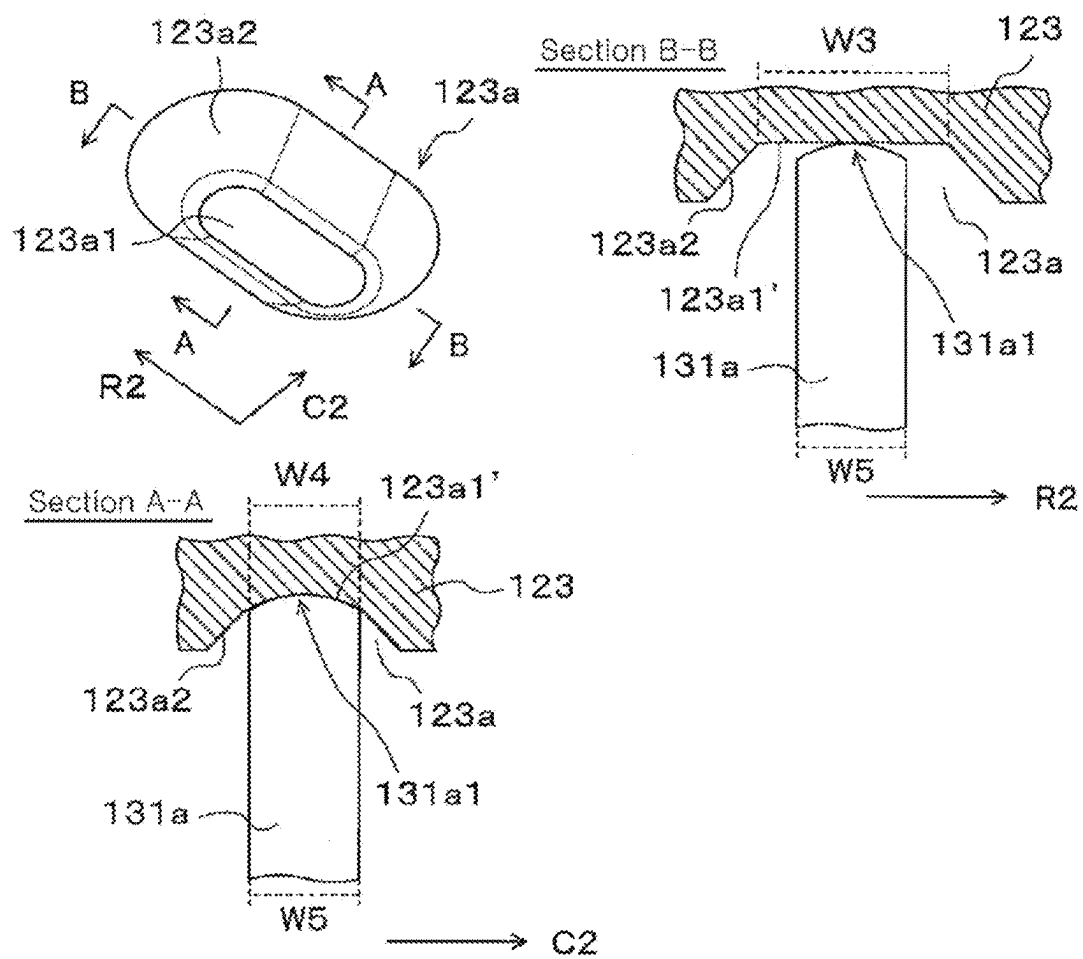
FIG. 5B

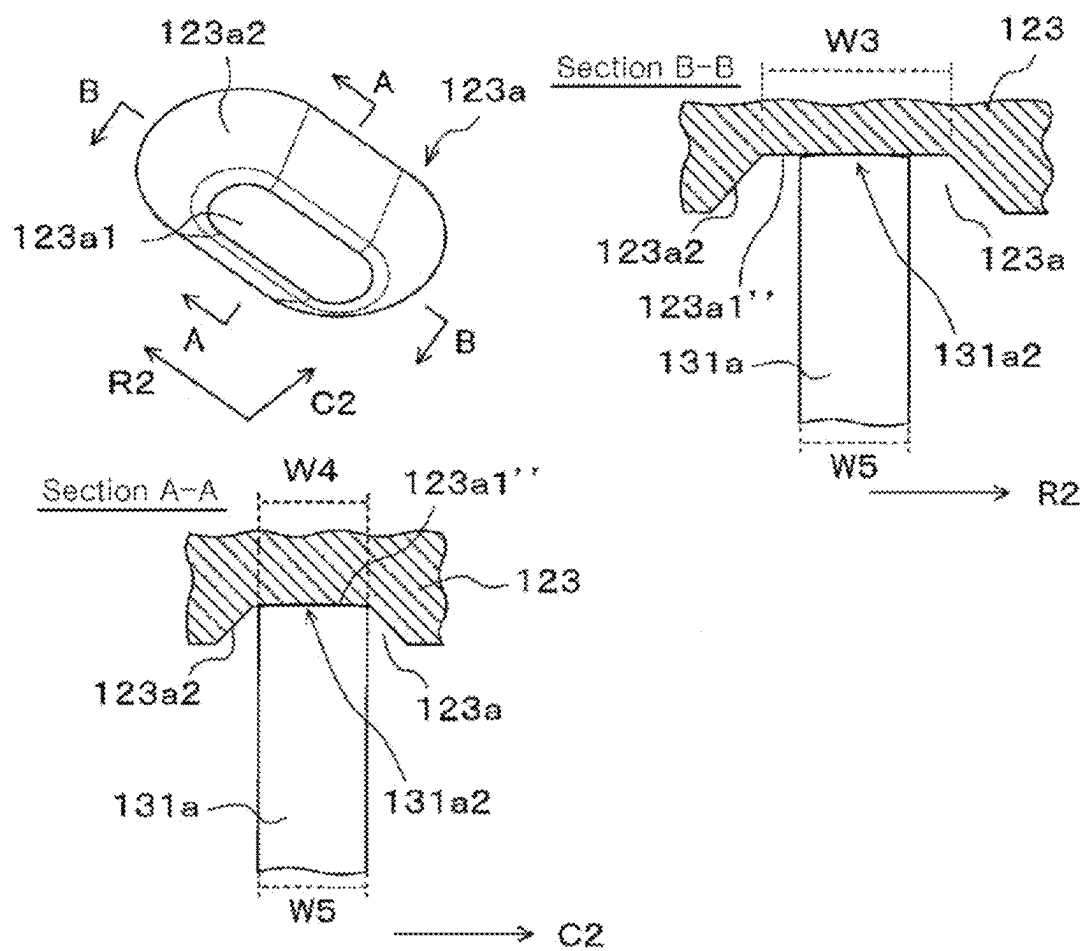
FIG. 6A

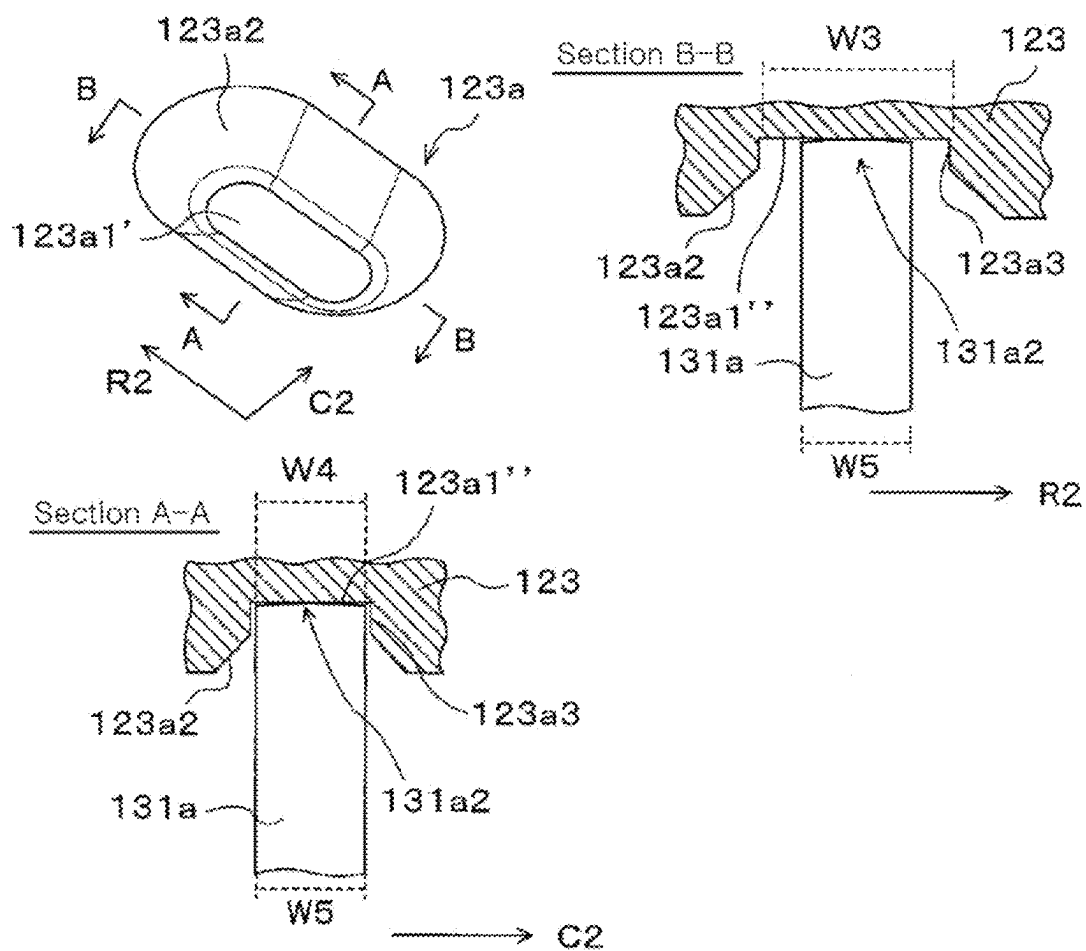
FIG. 6B

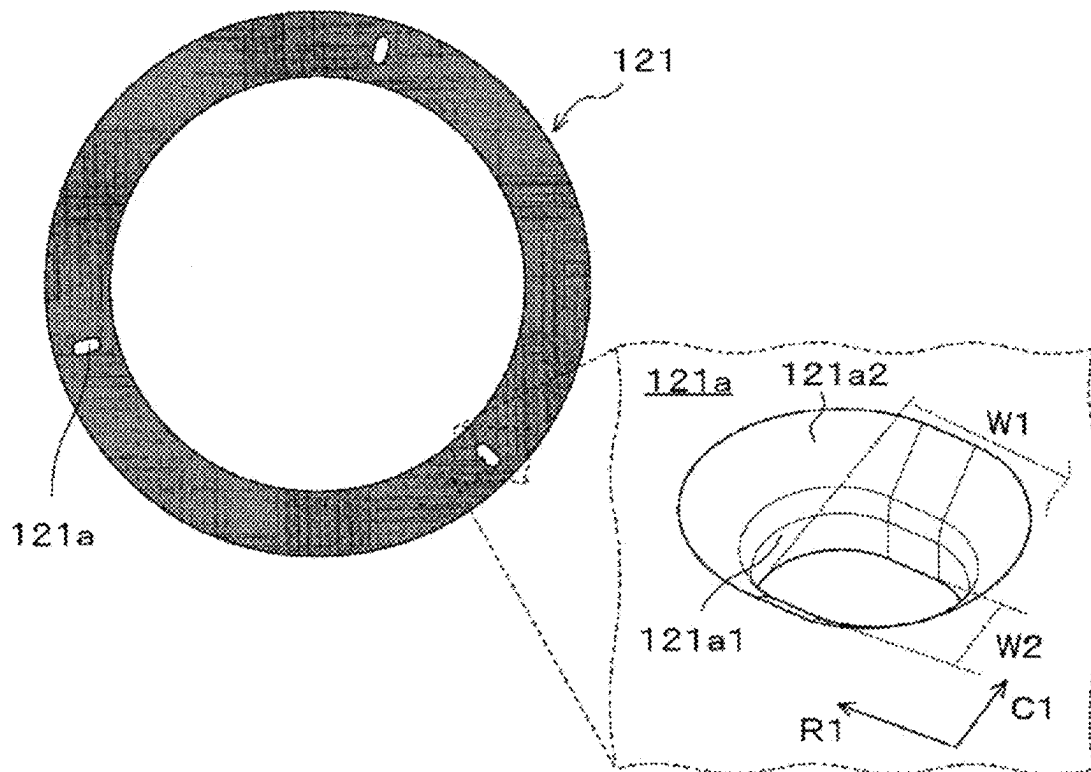
FIG. 6C

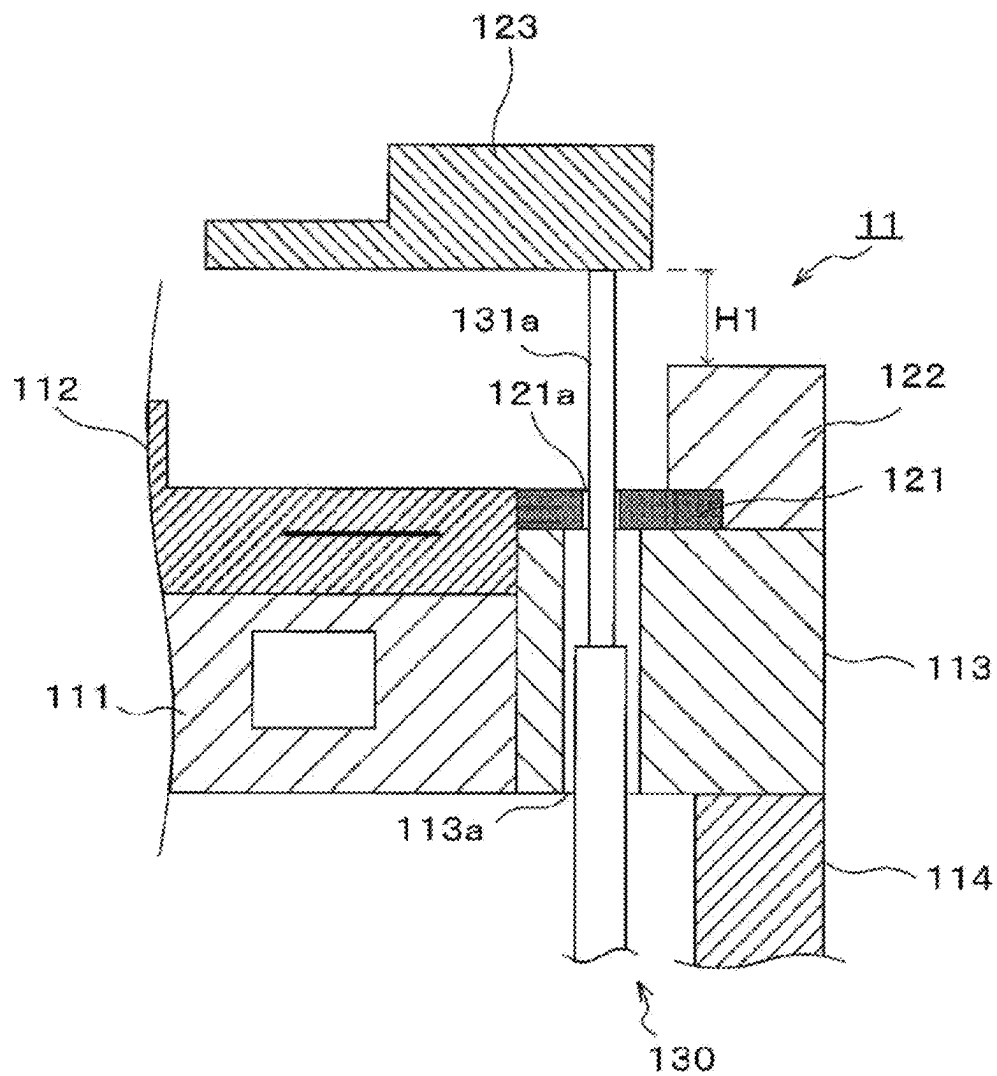
FIG. 7

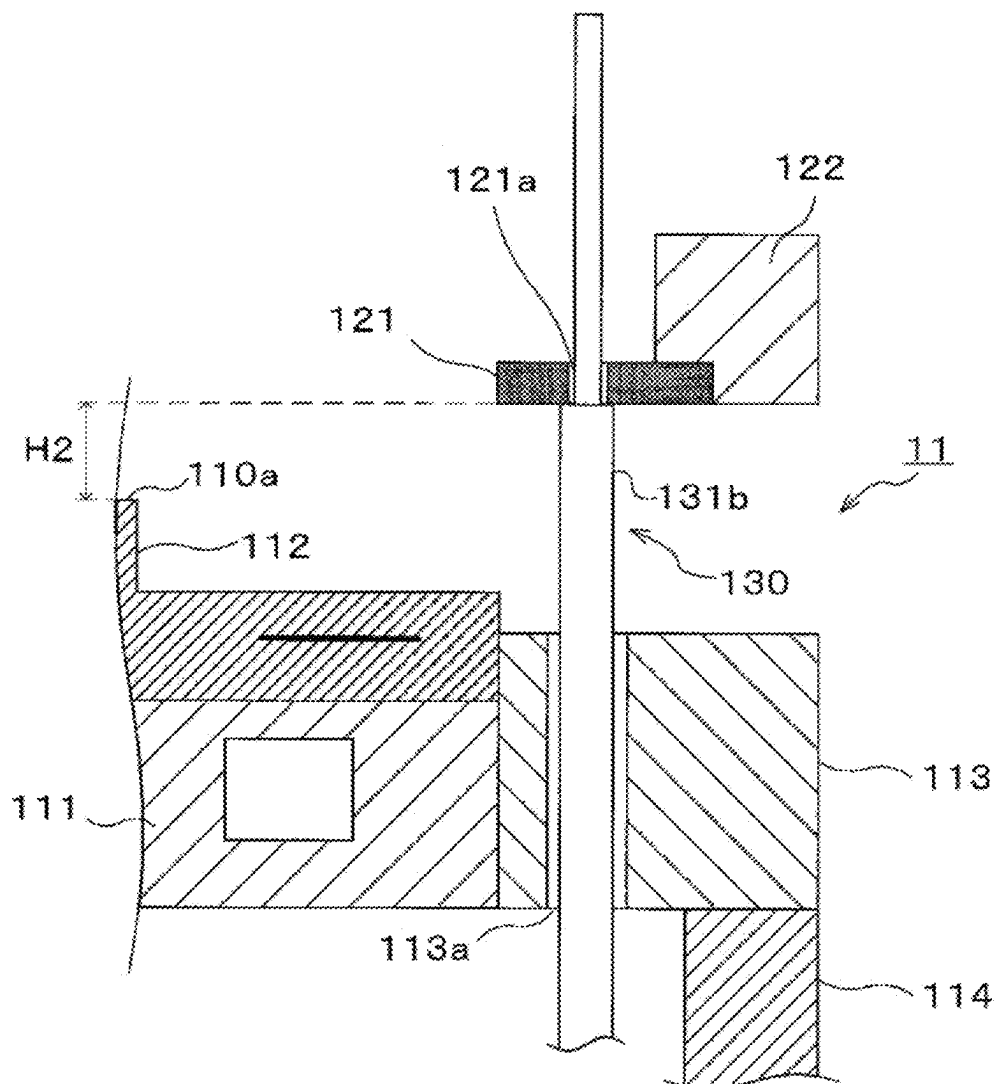
FIG. 8

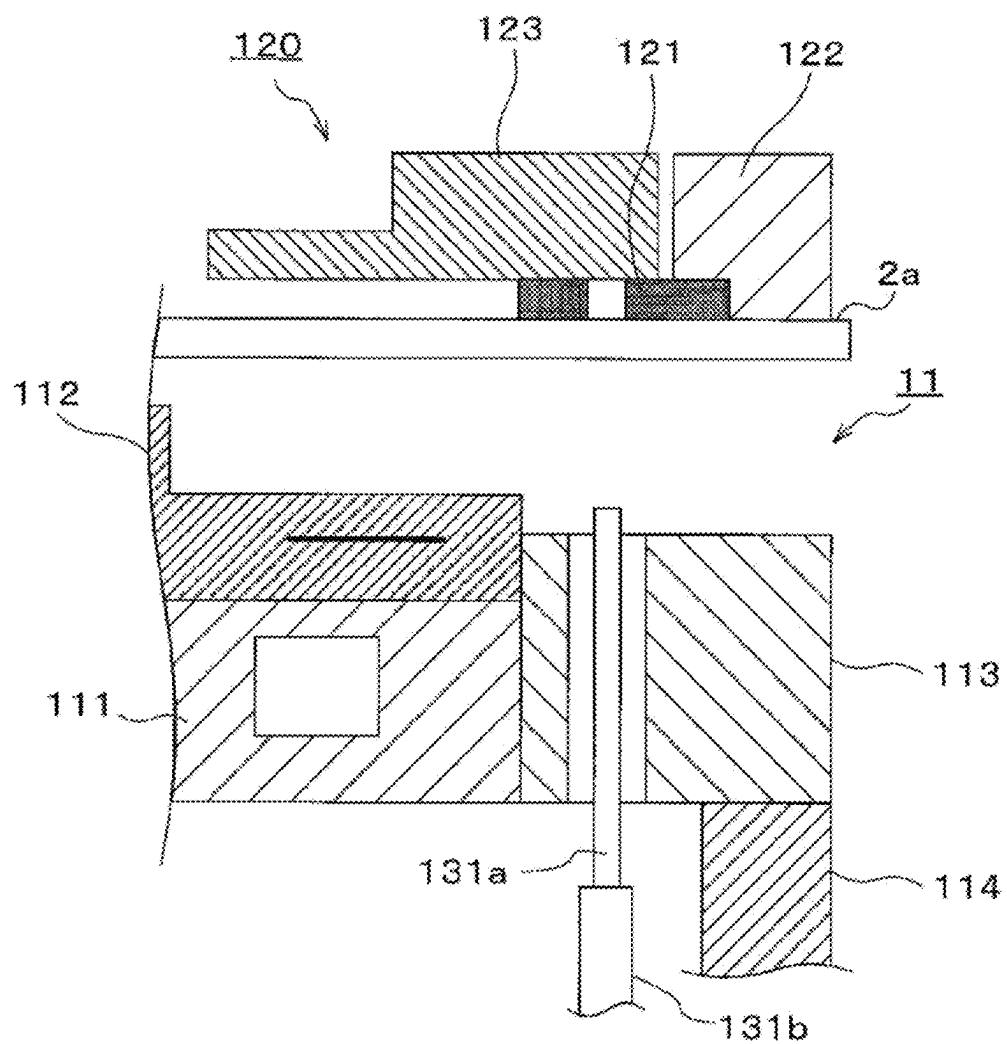
FIG. 9

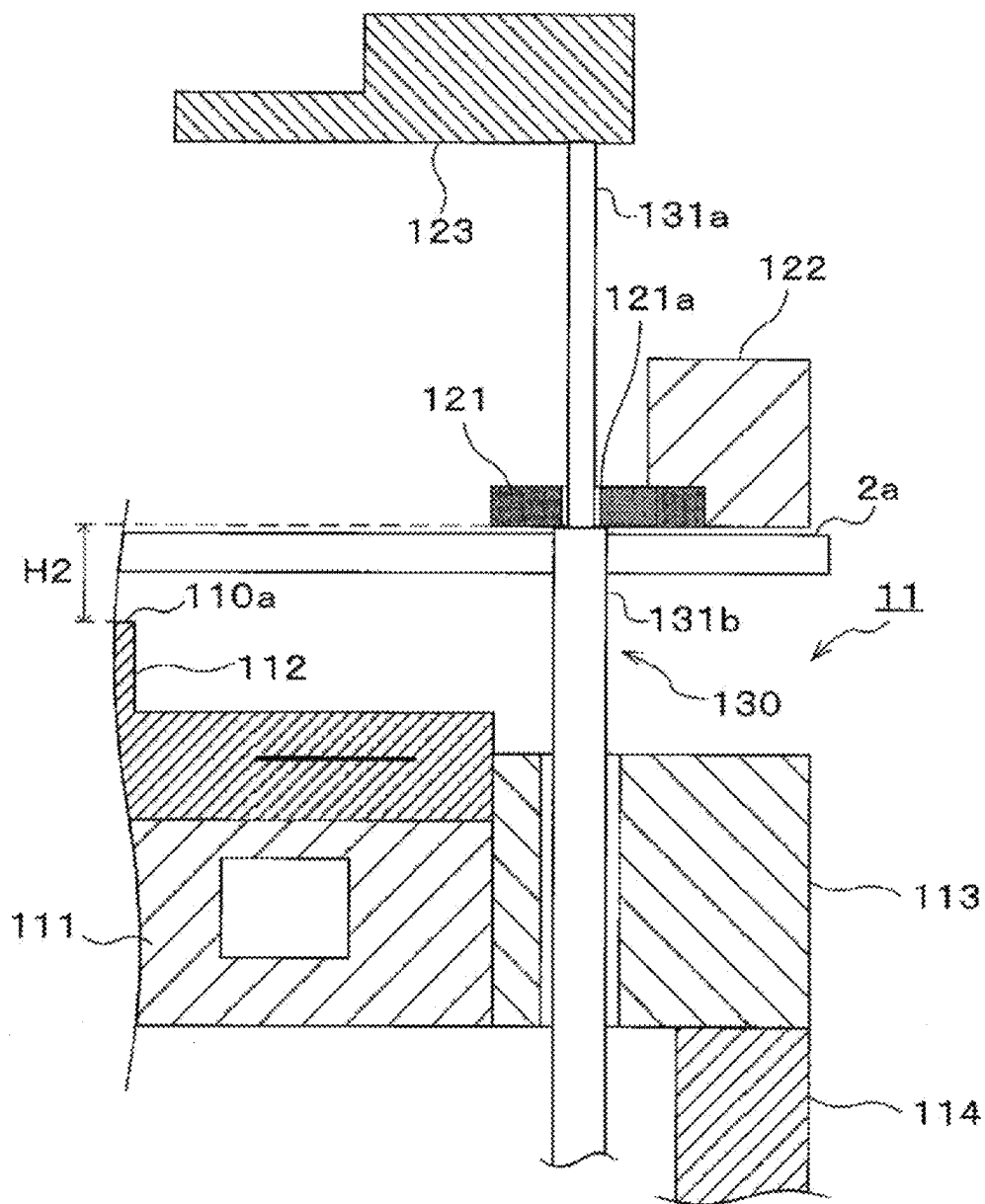
FIG. 10B

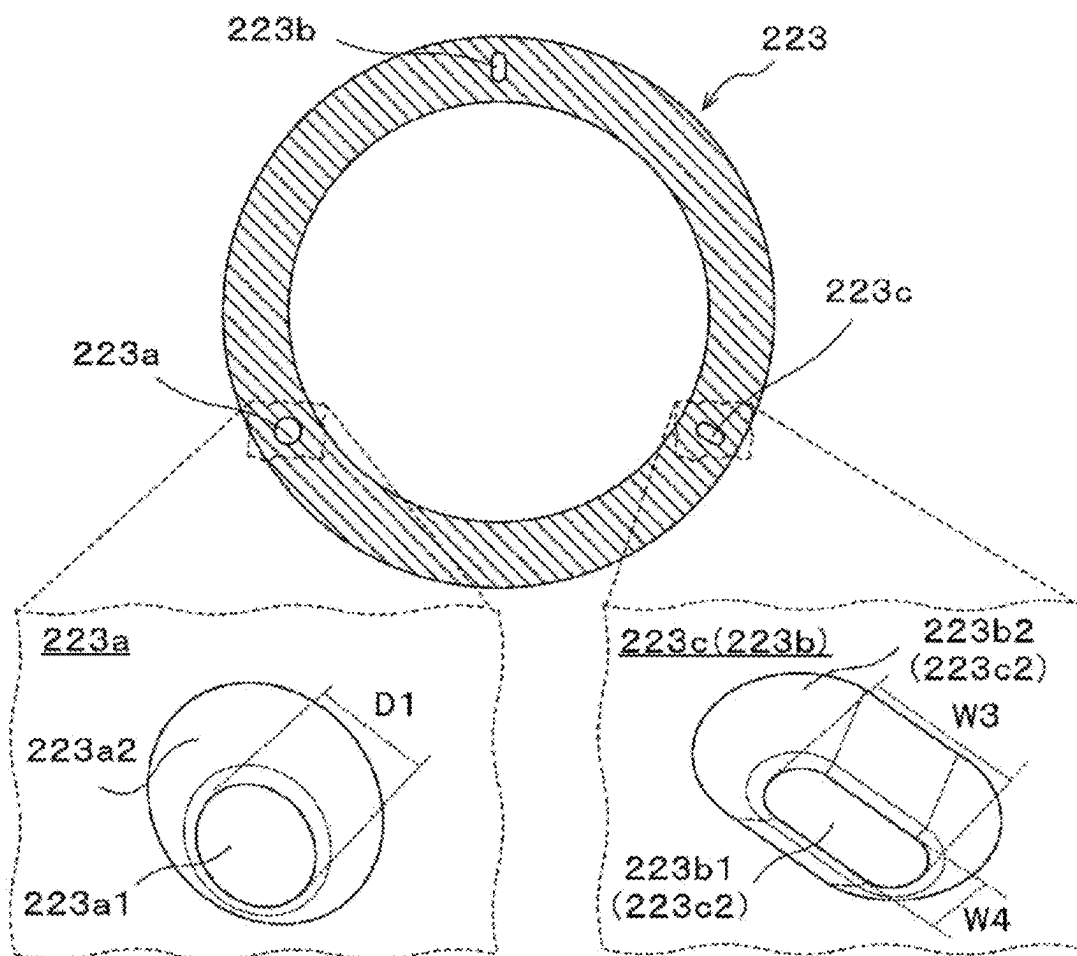
FIG. 11A

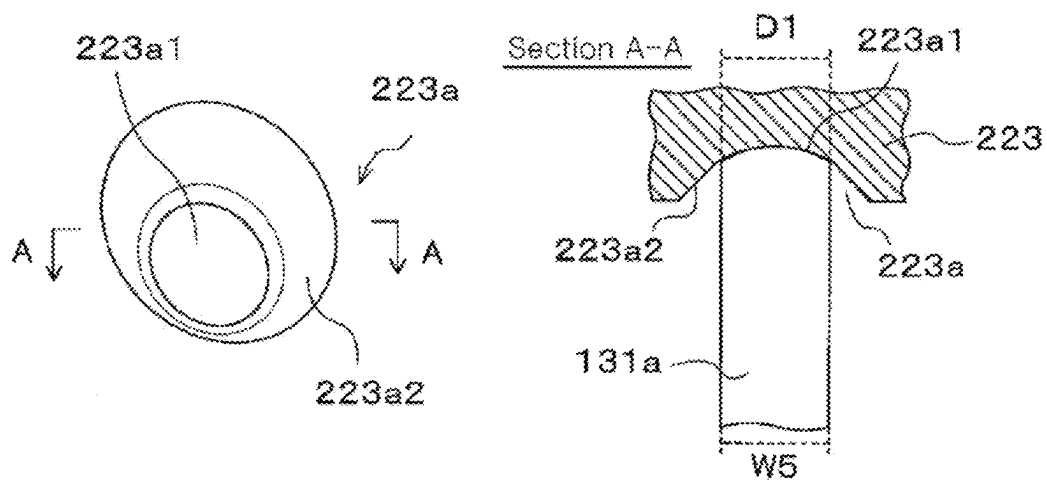
FIG. 11B

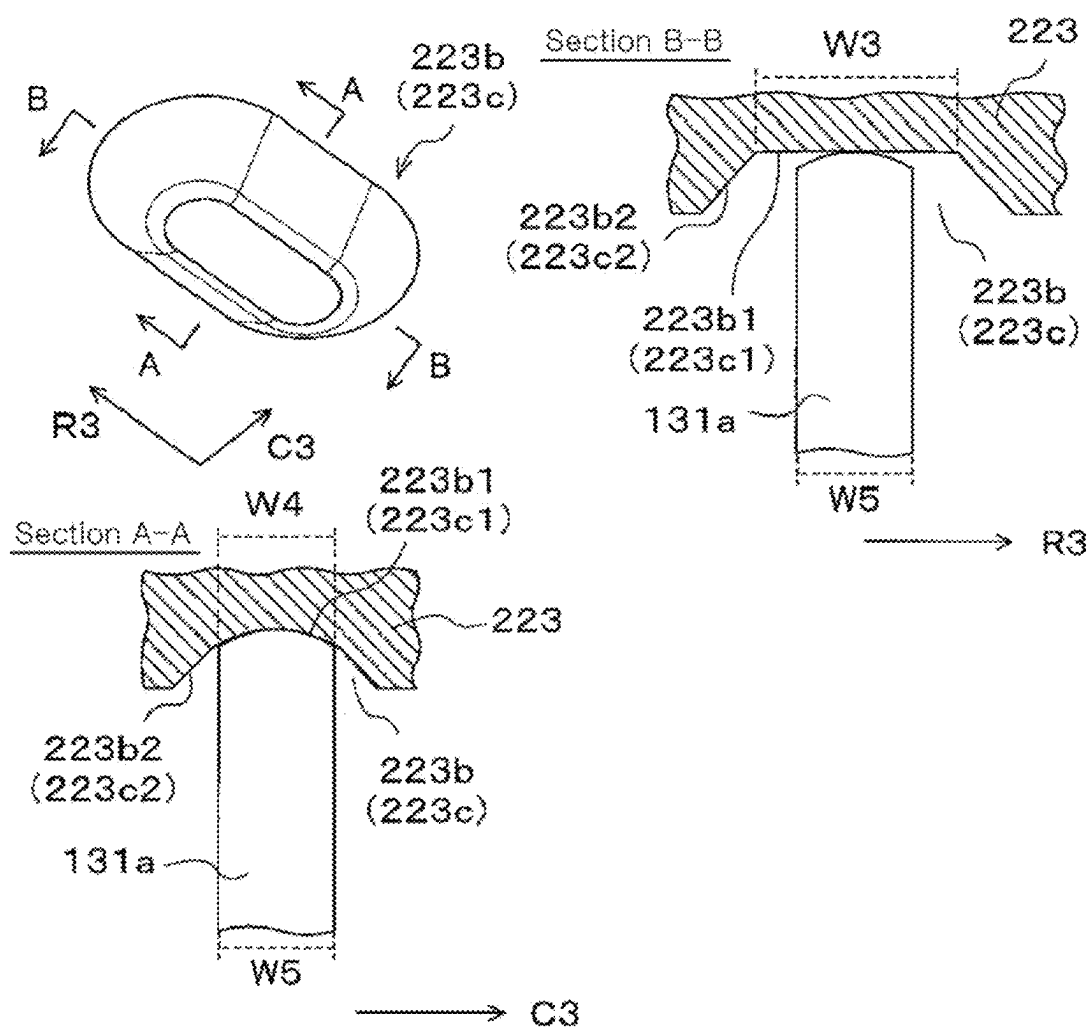
FIG. 11C

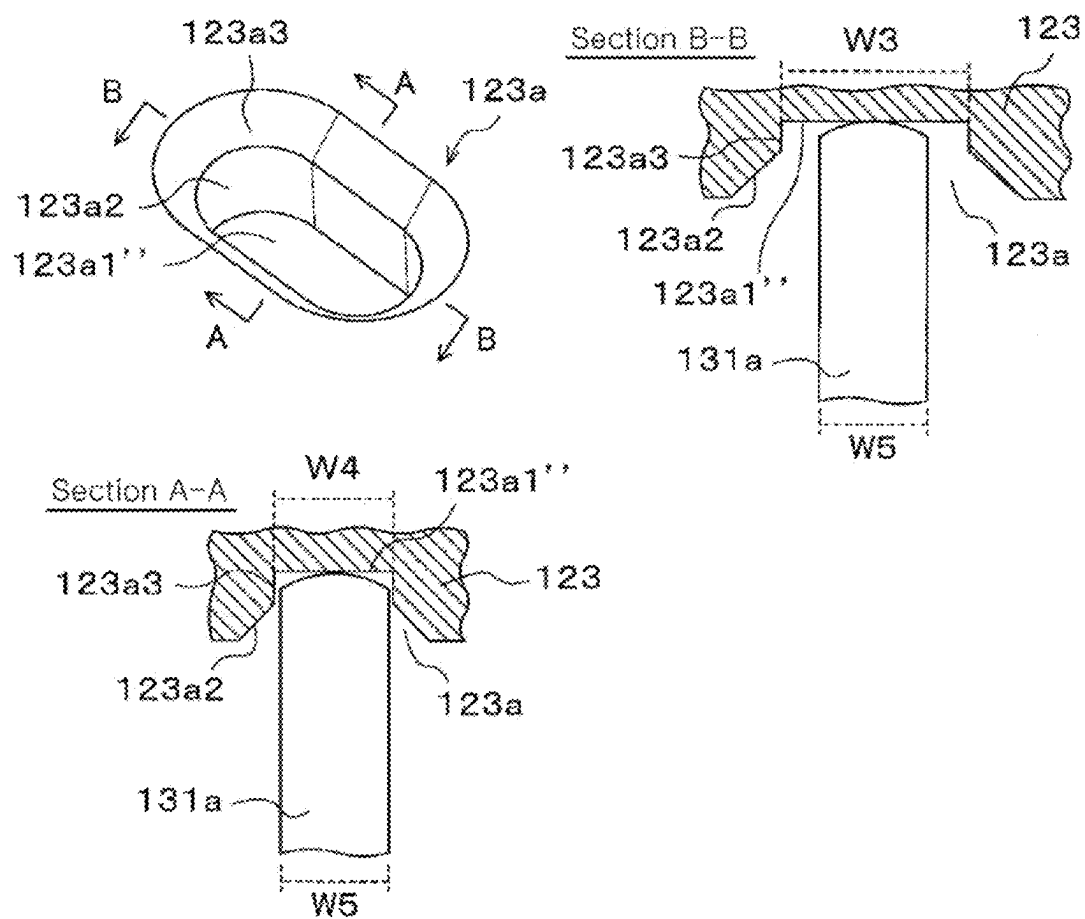
FIG. 12A

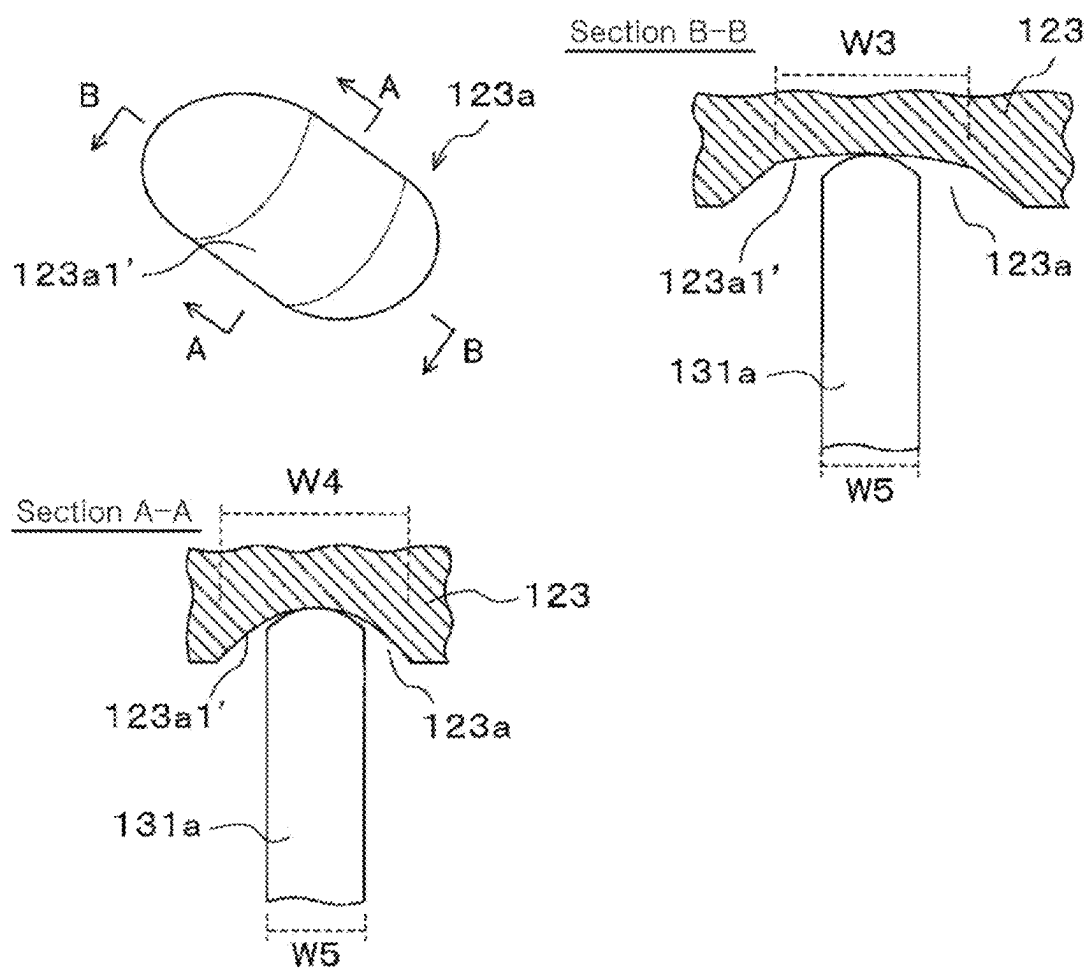
FIG. 12B

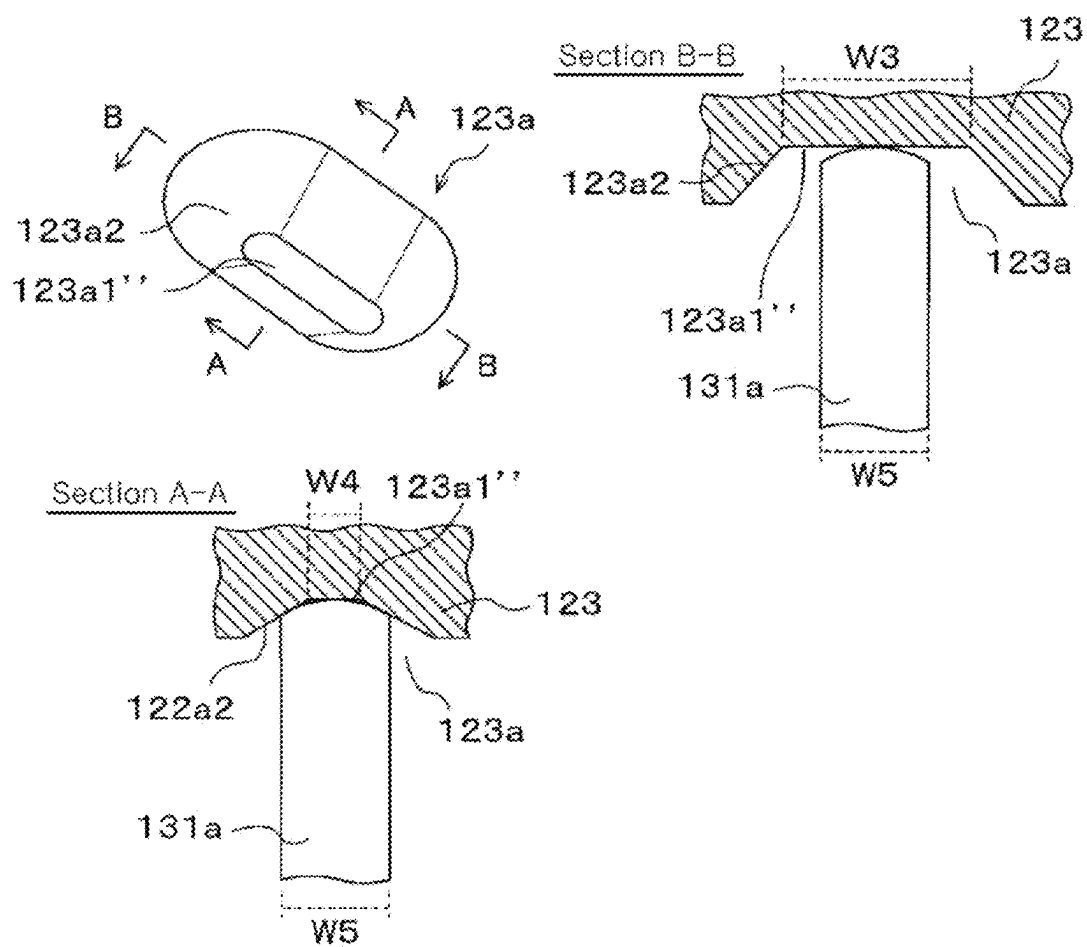
FIG. 12C

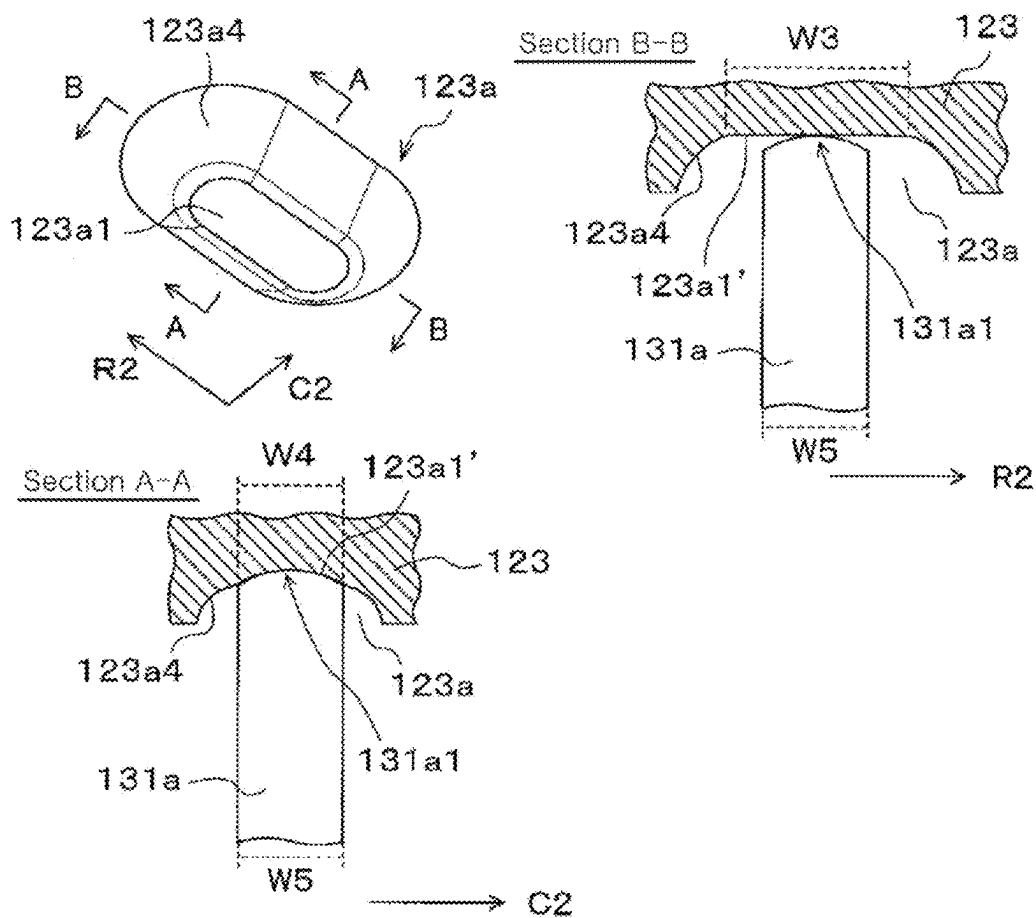
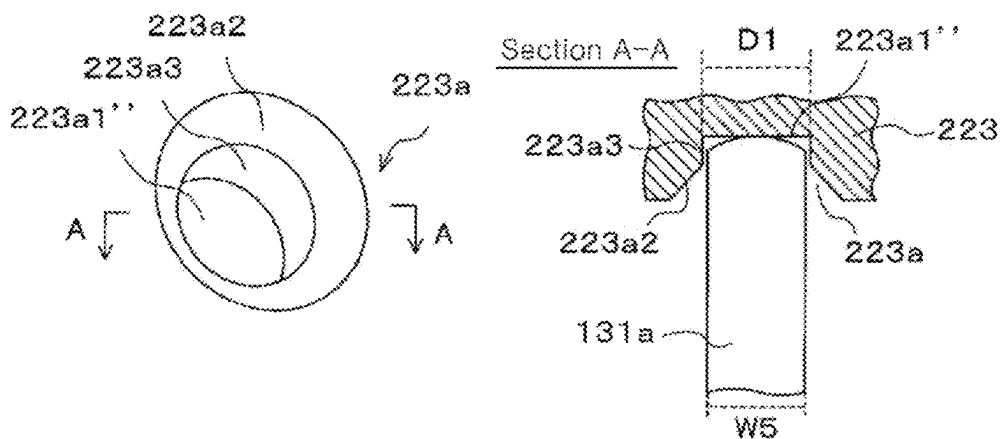
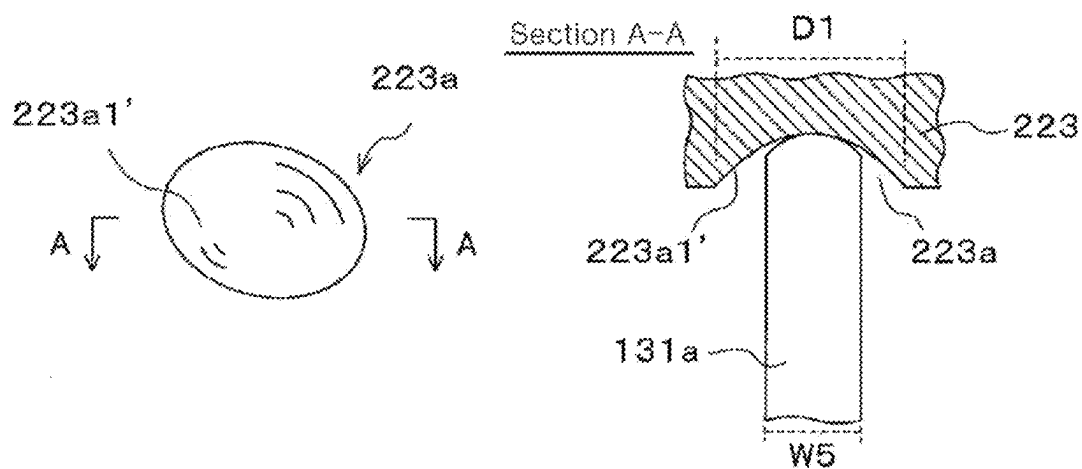
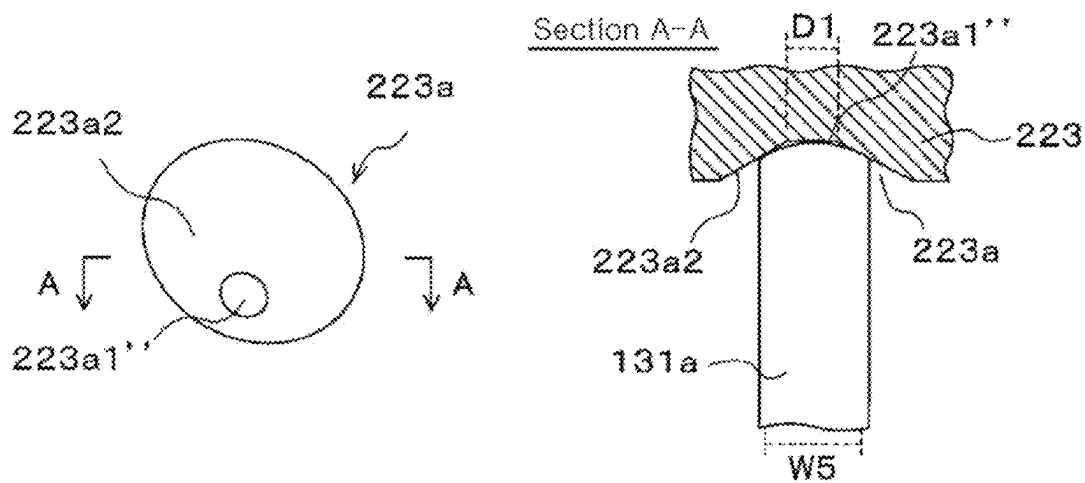
FIG. 12D*FIG. 13A*

FIG. 13B*FIG. 13C*

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PLASMA PROCESSING APPARATUS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application claims priority to Japanese Patent Application No. 2021-215091 filed on Dec. 28, 2021 and Japanese Patent Application No. 2022-163154 filed on Oct. 11, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a plasma processing apparatus.

BACKGROUND

Japanese Laid-open Patent Publication No. 2021-141313 discloses a substrate support having an annular member mounting surface on which an annular member is mounted, and three or more lifters which are configured to protrude from the annular member mounting surface. According to the description in the Japanese Laid-open Patent Publication, a concave portion formed by a concave surface which is concave upward is provided at a location on a bottom of the annular member corresponding to each lifter, and when viewed from a plane, the concave portion is larger than a transfer precision of the annular member upward of the annular member mounting surface, and is formed to be larger than an upper end portion of the lifter. Further, the upper end portion of the lifter is formed in a hemispherical shape, and the concave surface has a smaller curvature than the concave surface forming the hemispherical shape of the upper end portion of the lifter.

SUMMARY

The technique of the present disclosure provides a plasma processing apparatus that can properly position a ring assembly to a substrate support.

In accordance with an aspect of the present disclosure, there is a plasma processing apparatus, comprising: a plasma processing chamber; a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface; an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a radial direction of the insulating ring and a second width in a circumferential direction of the insulating ring, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side; a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a radial direction of the conductive ring and a fourth width in a circumferential direction of the conductive ring, the fourth width being smaller than the third width; at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting portion being configured to

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support the conductive ring from downward by contacting a bottom of the groove of the conductive ring through the through hole of the insulating ring, the lower supporting portion configured to support the insulating ring from downward by contacting an inclined surface of the insulating ring, the inclined surface defining the lower hole portion; and at least one actuator configured to vertically move the at least three lift pins.

BRIEF DESCRIPTION OF THE DRAWINGS

The objects and features of the present disclosure will become apparent from the following description of embodiments, given in conjunction with the accompanying drawings, in which:

FIG. 1 is an explanatory diagram illustrating an outline of a configuration of a plasma processing system;

FIG. 2 is a longitudinal cross-sectional view illustrating an outline of a configuration of a plasma processing apparatus according to an exemplary embodiment;

FIG. 3 is a partial enlarged view of FIG. 2;

FIG. 4A is an explanatory diagram illustrating an outline of a configuration of an insulating ring;

FIG. 4B is an explanatory diagram illustrating the outline of the configuration of the insulating ring;

FIG. 5A is an explanatory diagram illustrating the outline of the configuration of an inner edge ring;

FIG. 5B is an explanatory diagram illustrating the outline of the configuration of the inner edge ring;

FIG. 6A is an explanatory diagram illustrating another formation example of a groove for the inner edge ring;

FIG. 6B is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

FIG. 6C is an explanatory diagram illustrating another formation example of a through hole for the insulating ring;

FIG. 7 is a diagram illustrating a form of replacement processing of a ring assembly;

FIG. 8 is a diagram illustrating the form of the replacement processing of the ring assembly;

FIG. 9 is a diagram illustrating the form of the replacement processing of the ring assembly;

FIG. 10A is a diagram illustrating the form of the replacement processing of the ring assembly;

FIG. 10B is a diagram illustrating the form of the replacement processing of the ring assembly;

FIG. 11A is a diagram illustrating another configuration example of the inner edge ring;

FIG. 11B is a diagram illustrating another configuration example of the inner edge ring;

FIG. 11C is a diagram illustrating another configuration example of the inner edge ring;

FIG. 12A is an explanatory diagram illustrating another formation example for the groove for the inner edge ring;

FIG. 12B is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

FIG. 12C is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

FIG. 12D is an explanatory diagram illustrating another formation example of the groove for the inner edge ring;

FIG. 13A is an explanatory diagram illustrating another formation example for the groove for the inner edge ring;

FIG. 13B is an explanatory diagram illustrating another formation example of the groove for the inner edge ring; and

FIG. 13C is an explanatory diagram illustrating another formation example of the groove for the inner edge ring.

DETAILED DESCRIPTION

In a manufacturing process of semiconductor devices, plasma processing such as etching or the like is performed

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on a semiconductor substrate (hereinafter, simply referred to as "substrate") using plasma. The plasma processing is performed in a state in which a substrate is mounted on a substrate support placed in a processing container configured to be decompressed.

In addition, the substrate support includes a plurality of annular members placed to surround around the substrate on the substrate support in order to obtain an excellent and uniform processing result at a center and a periphery of the substrate at the time of the plasma processing. The plurality of annular members includes an edge ring placed adjacent to the substrate on the substrate support or a cover ring placed to cover an outer surface of the edge ring. Since the annular members are consumed by being exposed to plasma, the annular members need to be regularly replaced. The replacement of the annular member is executed, for example, by using a lifter which is lifted while supporting the annular member and a transfer mechanism transferring the annular member.

Here, when replacing the annular members using the lifter or transfer mechanism, the annular members may not be appropriately placed at desired positions on the substrate support due to transfer precision by the transfer mechanism, etc., for example. In addition, in recently implementing a transfer function of the annular member, enhancement of installation precision of the annular member on the substrate support is further require than the transfer precision by the transfer mechanism.

In the plasma processing apparatus disclosed in the Japanese Laid-open Patent Publication, a groove is formed on a bottom of an edge ring, the groove being larger than the transfer precision by the transfer mechanism, and a through hole is formed in the cover ring, the through hole being larger than the transfer precision by the transfer mechanism, thereby enhancing positioning precision of the annular member.

However, when the replacement of the annular member is executed by using the plasma processing apparatus described in the Japanese Laid-open Patent Publication, for example, there is a risk that the positioning precision of the annular member will deteriorate or damage the lifter or the annular member when a relative dislocation occurs between positions of the lifter and the groove (through hole) due to a mechanical difference (e.g., processing tolerance) or thermal expansion of the annular member.

The technique according to the present disclosure is derived by considering the circumstances, and provides a plasma processing apparatus that can properly position a ring assembly to the substrate support. Hereinafter, a configuration of a plasma processing system including a substrate processing apparatus according to the exemplary embodiment will be described with reference to the drawings. Further, in the present specification and drawings, the same reference numerals are given to elements having substantially the same functional configuration, so a redundant description will be omitted.

<Plasma Processing System>

In an exemplary embodiment, a plasma processing system includes a plasma processing apparatus 1, a transfer apparatus 2, and a controller 3, as illustrated in FIG. 1. The plasma processing system is an example of a substrate processing system, and the plasma processing apparatus 1 is an example of a substrate processing apparatus. The plasma processing apparatus 1 includes a plasma processing chamber 10, a substrate support 11, and a plasma generator 12. The plasma processing chamber 10 has a plasma processing space. Further, the plasma processing chamber 10 includes

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at least one gas inlet for supplying at least one processing gas to the plasma processing space and at least one gas outlet for exhausting gases from the plasma processing space. The gas inlet is connected to a gas supply 20 to be described below and the gas outlet is connected to an exhaust system 40 to be described below. The substrate support 11 is disposed in the plasma processing space, and has a substrate supporting surface for supporting a substrate.

The plasma generator 12 is configured to generate a plasma from at least one processing gas supplied in the plasma processing space. The plasma formed in the plasma processing space may be a capacitively coupled plasma (CCP), an inductively coupled plasma (ICP), an electron-cyclotron-resonance plasma (ECR plasma), a helicon wave plasma (HWP), or a surface plasma (SWP). Further, various types of plasma generator may be used, which include an alternating current (AC) plasma generator and a direct current (DC) plasma generator. In an exemplary embodiment, an AC signal (AC power) used in the AC plasma generator has a frequency in a range of 100 kHz to 10 GHz. Therefore, the AC signal includes a radio frequency (RF) signal and a microwave signal. In an exemplary embodiment, the RF signal has a frequency in a range of 100 kHz to 150 MHz.

The transfer apparatus 2 includes a transfer arm 2a that holds and transfers a substrate W and a ring assembly 120 to be described below. A wafer is an example of the substrate W. The transfer apparatus 2 is configured to transfer the substrate W and the ring assembly 120 between the outside of the plasma processing apparatus 1 and the substrate support 11 placed inside the plasma processing apparatus 1.

The controller 3 processes a computer executable command which allows the plasma processing apparatus 1 and the transfer apparatus 2 to execute various processes described in the present disclosure. The controller 3 may be configured to control each element of the plasma processing apparatus 1 and the transfer apparatus so as to execute various processes described herein. In an exemplary embodiment, a part or the entirety of the controller 3 may be included in the plasma processing apparatus 1. The controller 3 may include a processor 3a1, a storage 3a2, and a communication interface 3a3. The controller 3 may be implemented by a computer 3a, for example. The processor 3a1 may be configured to perform various control operations by reading a program from the storage 3a2 and executing the read program. The program may be stored in the storage 3a2 in advance or obtained through a medium as necessary. The obtained program is stored in the storage 3a2, and read and executed from the storage 3a2 by the processor 3a1. The medium may be various storage media readable by the computer 3a or a communication line connected to the communication interface 3a3. The processor 3a1 may be a central processing unit (CPU). The storage 3a2 may include a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or a combination thereof. The communication interface 3a3 may communicate with the plasma processing apparatus 1 through a communication line such as local area network (LAN), etc. Further, the storage media may be transitory or non-transitory.

<Plasma Processing Apparatus>

Next, as an example of the plasma processing apparatus 1, an example of a configuration of a capacitively coupled plasma processing apparatus 1 will be described. FIG. 2 is a longitudinal cross-sectional view illustrating an outline of a configuration of a plasma processing apparatus 1 according to an exemplary embodiment. FIG. 3 is a partial enlarged

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view enlarging and illustrating a part of a configuration of a substrate support 11 illustrated in FIG. 2.

The capacitively coupled plasma processing apparatus 1 includes a plasma processing chamber 10, a gas supply 20, a power supply 30, and an exhaust system 40. Further, the plasma processing apparatus 1 includes the substrate support 11 and a gas introduction unit. The gas introduction unit is configured to introduce at least one processing gas into the plasma processing chamber 10. The gas introduction unit includes a showerhead 13. The substrate support 11 is placed in the plasma processing chamber 10. The showerhead 13 is placed above the substrate support 11. In an exemplary embodiment, the showerhead 13 constitutes at least a part of a ceiling of the plasma processing chamber 10. The plasma processing chamber 10 has a plasma processing space 10s defined by the showerhead 13, a side wall 10a of the plasma processing chamber 10, and the substrate support 11. The plasma processing chamber 10 is grounded. The showerhead 13 and the substrate support 11 are electrically insulated from a housing of the plasma processing chamber 10.

The substrate support 11 includes a body 110, a ring assembly 120, and a lifter 130. The body 110 has a central region 110a for supporting the substrate W and an annular region 110b for supporting the ring assembly 120. The annular region 110b of the body 110 surrounds the central region 110a of the body 110 in a plan view. The substrate W is disposed on the central region 110a of the body 110, and the ring assembly 120 is placed on the annular region 110b of the body 110 to surround the substrate W on the central region 110a of the body 110. Therefore, the central region 110a is also called a substrate supporting surface for supporting the substrate W, and the annular region 110b is also called a ring supporting surface for supporting the ring assembly 120.

As illustrated in FIG. 3, in an exemplary embodiment, the body 110 includes a base 111, an electrostatic chuck 112, a support 113, and an insulator 114.

The base 111 includes a conductive member. The conductive member of the base 111 may serve as a lower electrode. The electrostatic chuck 112 is placed on the base 111. The electrostatic chuck 112 includes a ceramic member 112a and an electrostatic electrode 112b placed in the ceramic member 112a. The ceramic member 112a has the central region 110a. In an exemplary embodiment, the ceramic member 112a and the support 113 have the annular region 110b. Further, another member surrounding the electrostatic chuck 112 may have the annular region 110b such as an annular electrostatic chuck instead of an annular insulating member such as the support 113. The ring assembly 120 may be placed on the annular electrostatic chuck or the annular insulating member or placed on both the electrostatic chuck 112 and the annular insulating member. Alternatively, the constitution of the annular electrostatic chuck or the annular insulating member is omitted, and the ring assembly 120 may be placed only on the electrostatic chuck 112.

Further, at least one RF/DC electrode coupled to a radio frequency (RF) power supply 31 and/or direct current (DC) power supply 32 to be described below may be placed in the ceramic member 112a. In this case, at least one RF/DC electrode serves as the lower electrode. When a bias RF signal and/or DC signal to be described below is supplied to at least one RF/DC electrode, the RF/DC electrode is also called a bias electrode. Further, the conductive member of the base 111 and at least one RF/DC electrode may serve as a plurality of lower electrodes. Further, the electrostatic

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electrode 112b may serve as the lower electrode. Therefore, the substrate support 111 includes at least one lower electrode.

The support 113 is a member formed in a ring shape when viewed on the plan view by using an insulating material such as quartz, etc., for example and supports the base 111 and the electrostatic chuck 112. In an exemplary embodiment, the ring assembly 120 is mounted on the support 113 as described above. Further, a plurality of, in the exemplary embodiment, three through holes 113a penetrating in a thickness direction are formed in the support 113. A lift pin 131 to be described below of the lifter 130 is inserted into the through hole 113a as illustrated in FIG. 3.

The insulator 114 is a cylindrical member made of ceramic, etc., and supports the support 113. The insulator 114 is formed to have an outer diameter equal to an outer diameter of the support 113, for example, and supports a periphery of the support 113.

The ring assembly 120 as the annular member includes a plurality of annular members. As illustrated as an example in FIG. 3, the ring assembly 120 includes an insulating ring 121, an outer edge ring 122, and an inner edge ring 123, as the plurality of annular members. The outer edge ring 122 and the inner edge ring 123 are made of silicon, silicon carbide, or quartz, as an example. The insulating ring 121 is supported on the support 113 constituting the annular region 110b (ring supporting surface). The inner edge ring 123 is supported on the electrostatic chuck 112 and the insulating ring 121 constituting the annular region 110b (ring supporting surface). That is, the insulating ring 121 and the inner edge ring 123 are supported on the support 113 constituting the annular region 110b (ring supporting surface) in order. Further, the ring assembly 120 is lifted up from the annular region 110b (ring supporting surface) by an operation of the lifter 130, and thus configured to be transferred by the transfer apparatus 2 placed outside the plasma processing apparatus 1.

Further, the inner edge ring 123 may constitute a “conductive ring” according to the technique of the present disclosure. Further, the outer edge ring 122 may constitute an “additional conductive ring” according to the technique of the present disclosure. In other words, the ring assembly 120 as the annular member includes an insulating ring 121, an additional conductive ring 122, and a conductive ring 123 as a plurality of annular members.

The outer edge ring 122 is formed large in the thickness direction as compared with the insulating ring 121, and placed to surround the inner edge ring 123 and the insulating ring 121, and placed so that an inner part of the outer edge ring 122 overlaps with an outer part of the insulating ring 121 in a vertical direction. In other words, an upper surface of the insulating ring 121 constitutes a support surface for the outer edge ring 122 and the inner edge ring 123.

Further, the configuration of the ring assembly 120 is not limited to an illustrated example, and for example, the insulating ring 121 and the outer edge ring 122 may be integrally configured. In this case, the insulating ring 121 and the outer edge ring 122 may be configured by the same member, that is, the outer edge ring 122 may serve as a part of the insulating ring and the insulating ring may serve as a part of the additional conductive ring.

Further, a plurality of, in the exemplary embodiment, three through holes 121a penetrating in the thickness direction are formed in the insulating ring 121 as illustrated in FIGS. 4A and 4B. The through hole 121 is formed at a position corresponding to the through hole 113a formed in

the support **113**, and an upper supporting portion **131a** of the lift pin **131** to be described below is inserted into the through hole **121a**.

Three through hole **121a** has an upper hole portion **121a1** and a lower hole portion **121a2**. The upper hole portion **121a1** of the through hole **121a** has a substantially rectangular shape in which a first width **W1** which extends in a radial direction **R1** of the insulating ring **121** is formed to be larger than a second width **W2** which extends in a circumferential direction **C1** of the insulating ring **121** orthogonal to the radial direction **R1**. The first width **W1** is formed to be larger than at least a diameter (a fifth width **W5** to be described below) of the upper supporting portion **131a** of the lift pin **131** to be described below. The second width **W2** is formed to be larger than the diameter (fifth width **W5**) of the upper supporting portion **131a**, and further, formed to be smaller than a diameter (sixth width **W6** to be described below) of a lower supporting portion **131b** of the lift pin **131** to be described below.

The lower hole portion **121a2** of the insulating ring **121** has a flare shape (inclined surface **1211a**) in which the first width **W1** and the second width **W2** of the upper hole portion **121a1** spread downward (toward a bottom of the insulating ring **121**). Preferably, a lower end of the inclined surface **1211a** of the lower hole portion **121a2** is formed to be at least larger than the transfer precision of the ring assembly **120** by the transfer apparatus **2**.

In the insulating ring **121** according to the exemplary embodiment, therefore, the upper supporting portion **131a** of the lift pin **131** is inserted into the upper hole portion **121a1**, and the lower supporting portion **131b** of the lift pin **131** comes into contact with two points on the inclined surface **1211a** of the lower hole portion **121a2** in the circumferential direction **C1**.

Further, as described above, in the upper hole portion **121a1**, the first width **W1** which extends in the radial direction **R1** is formed to be larger than the second width **W2** which extends in the circumferential direction **C1**, but the sizes of the first width **W1** and the second width **W2** are preferably set to the same as each other as possible. In other words, the upper hole portion **121a1** preferably has a substantially square shape when viewed in the plan view.

The inner edge ring **123** as the conductive ring may also be called a focus ring, and enhances in-plane uniformity of the plasma processing for the substrate **W**. Further, the outer edge ring **122** as the additional conductive ring substantially expands an area of a conductive part (inner edge ring **123**) viewed in the plane on the substrate support **11**. The inner edge ring **123** and the outer edge ring **122** may be made of the silicon, the silicon carbide, or the quartz, as an example.

A plurality of, in the exemplary embodiment, three grooves **123a** are formed on the bottom of the inner edge ring **123** as illustrated in FIGS. **5A** and **5B**. The groove **123a** is formed at a position corresponding to the through hole **121a** formed in the insulating ring **121**, and a tip of the upper supporting portion **131a** of the lift pin **131** inserted into the through hole **121a** contacts the bottom **123a1**.

Three grooves **123a** have a substantially long hole shape in which a third width **W3** which extends in a radial direction **R2** of the inner edge ring **123** is formed to be larger than a fourth width **W4** which extends in a circumferential direction **C2** of the inner edge ring **123** orthogonal to the radial direction **R2**. The third width **W3** is formed to be larger than a diameter (a fifth width **W5**) of the upper supporting portion **131a**. The fourth width **W4** is formed to have the substantially same size as the diameter (the fifth width **W5** to be

described below) of the upper supporting portion **131a** of the lift pin **131** to be described below.

Further, the bottom **123a1** of three grooves **123a** has a shape of being fitted into a tip shape of the upper supporting portion **131a** of the lift pin **131**, in particular, in an extension direction of the fourth width **W4**. Specifically, in the exemplary embodiment, as described below, the upper supporting portion **131a** has a hemispherical shape **131a1** in which the tip protrudes as described below, but the bottom **123a1** of the groove **123a** is formed in a concave shape (hereinafter, the bottom **123a1** having a curve shape may be referred to as a "curved bottom **123a1**") having the substantially same curvature as the hemispherical shape **131a1**.

An inclined portion is formed on a flare-shaped side surface **123a2** of three grooves **123a** so that the third width **W3** and the fourth width **W4** spread downward (toward the bottom of the inner edge ring **123**). A lower end of the inclined surface **1211a** of the groove **123a** is preferably formed to be at least larger than the transfer precision of the inner edge ring **123** by the transfer apparatus **2**.

In the inner edge ring **123** according to the exemplary embodiment, therefore, the upper supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** in a circumferential direction (a fourth width (**W4**) direction) of the inner edge ring **123**.

Further, a depth of the groove **123a** is 0.4 mm or more and 1.0 mm or less, as an example.

In an exemplary embodiment, the lifter **130** includes a plurality of, in the exemplary embodiment, three lift pins **131** corresponding to the through holes **121a** of the insulating ring **121** and the grooves **123a** of the inner edge ring **123**, and at least one actuator **132**. Further, the lift pin **131** includes a plurality of cylinder supporting portions having different diameters. As illustrated as an example in FIG. **3**, the lift pin **131** includes the upper supporting portion **131a** and the lower supporting portion **131b**. The upper supporting portion **131a** and the lower supporting portion **131b** may be integrally configured.

The upper supporting portion **131a** at least has the fifth width **W5** having a diameter smaller than the first width **W1** and the second width **W2** of the through hole **121a** formed in the insulating ring **121**. The fifth width **W5** has the substantially same size as the fourth width **W4** of the groove **123a** formed in the inner edge ring **123**. The upper supporting portion **131a** is concatenated in an axial direction from the upper surface of the lower supporting portion **131b** to be described below, and moves in a vertical direction (axial direction) integrally with the lower supporting portion **131b** by the operation of the actuator **132**. In addition, the upper supporting portion **131a** is configured to be projected and depressed from the upper surface of the insulating ring **121** via the through hole **121a**, and therefore, supports the bottom of the inner edge ring **123** supported on the upper surface of the insulating ring **121**, more specifically, the bottom **123a1** of the groove **123a** formed in the inner edge ring **123**, and moves (lifts up) the bottom **123a1** in the vertical direction.

Further, the tip of the upper supporting portion **131a** is formed in a hemispherical shape **131a1** which gradually becomes thinner upward (see FIG. **5B**). The hemispherical shape **131a1** has the substantially same curvature as a concave surface of a curved bottom **123a1** of the groove **123a** formed in the inner edge ring **123**. Therefore, the tip of the upper supporting portion **131a** is configured to be in line-contact with the groove **123a** formed in the inner edge ring **123**.

The lower supporting portion **131b** at least has a diameter (the sixth width **W6** of FIG. 3) larger than the second width **W2** of the through hole **121a** formed in the insulating ring **121**. That is, the lower supporting portion **131b** has a step which protrudes outward in the radial direction from the outer periphery of the upper supporting portion **131a**, on the top surface thereof. In addition, the lower supporting portion **131b** is configured to support the inclined surface **1211a** formed in the lower hole portion **121a2** of the through hole **121a**, and therefore, supports the bottom of the insulating ring **121** and moves (lifts up) the bottom in the vertical direction. Further, in this case, the outer edge ring **122** held on the insulating ring **121** is also moved (lifted up) in the vertical direction at the same time.

The actuator **132** moves the lift pin **131** in the vertical direction (axial direction) to lift the ring assembly **120** (insulating ring **121**, the outer edge ring **122**, and the inner edge ring **123**) above the annular region **110b** (ring supporting surface). Therefore, the ring assembly **120** is transferred between the substrate support **11** and the transfer arm **2a** of the transfer apparatus **2**. One example of the actuator includes an electrical actuator or an air cylinder, a motor, etc.

Further, the number of actuators **132** placed in the lifter **130** is not particularly limited. That is, for example, the plurality of lift pins **131** may be integrally moved in the vertical direction by one actuator **132**. Further, for example, a plurality of actuators **132** may be placed to correspond to the lift pins **131**, respectively, and the respective lift pins **131** may be independently moved in the vertical direction.

Further, the substrate support **11** may include a temperature control module configured to control a temperature of at least one of the electrostatic chuck **112**, the ring assembly **120**, and the substrate **W** to a target temperature. The temperature control module may include a heater, a heat transfer medium, a path **111a**, or a combination thereof. In the flow path **111a**, a heat transfer fluid such as brine or gas flows. In an exemplary embodiment, the flow path **111a** is formed in the base **111** and one or more heaters are disposed in the ceramic member **112a** of the electrostatic chuck **112**. Further, the substrate support **11** may include a heat transfer gas supply configured to supply heat transfer gas (backside gas) to a gap between a back surface of the substrate **W** and the upper surface of the electrostatic chuck **112**.

FIG. 2 is described again.

The showerhead **13** is configured to introduce at least one processing gas from the gas supply **20** into the plasma processing space **10s**. The showerhead **13** includes at least one gas inlet **13a**, at least one gas diffusion space **13b**, and a plurality of gas introduction ports **13c**. The processing gas supplied from the gas supply **20** to the gas inlet **13a** is introduced into the plasma processing space **10s** from the plurality of gas introduction ports **13c** by passing through the gas diffusion space **13b**. Further, the showerhead **13** includes at least one upper electrode. Further, the gas introduction unit may include one or more side gas injectors (SGI) mounted on one or more openings formed on the side wall **10a** in addition to the showerhead **13**.

The gas supply **20** may include at least one gas source **21** and at least one flow controllers **22**. In an exemplary embodiment, the gas supply **20** is configured to supply at least one processing gas to the showerhead **13** from the gas sources **21** corresponding to the one or more processing gas, respectively through the flow controllers **22** corresponding thereto, respectively. Each flow controller **22** may include, for example, a mass flow controller or a pressure control type flow controller. Further, the gas supply **20** may include

at least one flow modulation device which modulates or pulses the flow of at least one processing gas.

The power supply **30** includes an RF power supply **31** coupled to the plasma processing chamber **10** through at least one impedance matching circuit. The RF power supply **31** is configured to supply at least one RF signal (RF power) to at least one lower electrode and/or at least one upper electrode. Therefore, plasma is generated from at least one processing gas supplied to the plasma processing space **10s**. Therefore, the RF power supply **31** may serve as at least a part of the plasma generator **12**. Further, by supplying the bias RF signal to at least one lower electrode, a bias potential may be generated on the substrate **W** and ion components in the formed plasma may be attracted to the substrate **W**.

In an exemplary embodiment, the RF power supply **31** includes a first RF generator **31a** and a second RF generator **31b**. The first RF generator **31a** is configured to be coupled to at least one lower electrode and/or at least one upper electrode through at least one impedance matching circuit, and to generate a source RF signal (source RF power) for plasma generation. In an exemplary embodiment, the source RF signal has a frequency in a range of 10 MHz to 150 MHz. In an exemplary embodiment, the first RF generator **31a** may be configured to generate a plurality of RF signals having different frequencies. One or more generated source RF signals are supplied to at least one lower electrode and/or at least one upper electrode.

The second RF generator **31b** is configured to be coupled to at least one lower electrode through at least one impedance matching circuit, and to generate the bias RF signal (bias RF power). The frequency of the bias RF signal may be the same as or different from the frequency of the source RF signal. In an exemplary embodiment, the bias RF signal has a frequency lower than the source RF signal. In an exemplary embodiment, the source RF signal has a frequency in a range of 100 kHz to 60 MHz. In an exemplary embodiment, the second RF generator **31b** may be configured to generate a plurality of bias RF signals having different frequencies. One or more generated bias RF signals are supplied to at least one lower electrode. Further, in various exemplary embodiments, at least one of the source RF signal and the bias RF signal may be pulsed.

Further, the power supply **30** may include the DC power supply **32** coupled to the plasma processing chamber **10**. The DC power supply **32** includes a first DC generator **32a** and a second DC generator **32b**. In an exemplary embodiment, the first DC generator **32a** is configured to be connected to at least one lower electrode and generate a first DC signal. The generated first DC signal is applied to at least one lower electrode. In an exemplary embodiment, the second DC generator **32b** is configured to be connected to at least one upper electrode and generate a second DC signal. The generated second DC signal is applied to at least one upper electrode.

In various exemplary embodiments, the first and second DC signals may be pulsed. In this case, a sequence of voltage pulses is applied to at least one lower electrode and/or at least one upper electrode. The voltage pulse may have a pulse waveform of a rectangle, a trapezoid, a triangle, or a combination thereof. In an exemplary embodiment, a waveform generator for generating a sequence of voltage pulses from the DC signal is connected between the first DC generator **32a** and at least one lower electrode. Therefore, the first DC generator **32a** and the waveform generator constitute a voltage pulse generator. When the second DC generator **32b** and the waveform generator constitute the voltage pulse generator, the voltage pulse generator is con-

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nected to at least one upper electrode. The voltage pulse may have a positive polarity and have a negative polarity. Further, the sequence of voltage pulses may include one or more positive voltage pulses and one or more negative voltage pulses within one cycle. Further, the first and second DC generators 32a and 32b may be provided in addition to the RF power supply 31, or the first DC generator 32a may be provided instead of the second RF generator 31b.

The exhaust system 40 may be connected to a gas outlet 10e provided on a bottom of the plasma processing chamber 10, for example. The exhaust system 40 may include a pressure adjustment valve and a vacuum pump. By the pressure adjustment valve, a pressure in the plasma processing space 10s is adjusted. The vacuum pump may include a turbo molecule pump, a dry pump, or a combination thereof.

Hereinafter, various exemplary embodiments have been described, but the present disclosure is not limited to the exemplary embodiment, but various additions, omissions, substitutions, and modifications may be made. Further, it is possible to form another exemplary embodiment by combining elements in different exemplary embodiments.

For example, in the exemplary embodiment, the tip of the upper supporting portion is formed in the hemispherical shape 131a1 as illustrated in FIG. 3, but the tip of the upper supporting portion 131a may substantially have a flat surface 131a2 as illustrated in FIG. 6A. In this case, in order to contact the tip of the upper supporting portion 131a and the bottom of the groove 123a, the bottom shape of the groove 123a also preferably has a flat bottom 123a1' which engages with the tip of the upper supporting portion 131a instead of the curved bottom 123a1'.

Further, in this case, as illustrated in FIG. 6B, at least three grooves 123a of the conductive ring 123 may be defined by the flat bottom 123a1', a vertical side surface 123a3 which extends downward from the flat bottom 123a1', and a flare-shaped side surface 123a2 which spreads downward from the vertical side surface 123a3.

Further, for example, in the exemplary embodiment, the through hole 121a formed in the insulating ring 121 is formed in the substantially rectangular shape, and preferably formed in the substantially square shape, but as illustrated in FIG. 6C, the through hole 121a may be formed in the substantially long hole shape in which a round shape of four corners is large. In this case, the through hole 121a having the substantially long hole shape preferably has the long hole shape which has the first width W1 which extends in the radial direction R1 of the insulating ring 121 is larger than the second width W2 which extends in the circumferential direction C1, i.e., has a long-side direction as the radial direction R1 of the insulating ring 121.

<Effects, Etc.>

The plasma processing system according to an exemplary embodiment is configured as above. That is, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber 10, a substrate support 11, an insulating ring 121, a conductive ring 123, at least three lift pins 131, and at least one actuator 132. The substrate support 11 is disposed in the plasma processing chamber 10, and has a substrate supporting surface 110a and a ring supporting surface 110b. The insulating ring 121 is placed on the ring supporting surface 110b, and has at least three through holes 121a. Each of at least three through holes 121a has an upper hole portion 121a1 and a lower hole portion 121a2. The upper hole portion 121a1 has the first width W1 in the radial direction R1 of the insulating ring 121 and the second width W2 in the circumferential direction C1 of the insulating ring 121. The second width W2 is

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smaller than the first width W1, and the lower hole portion 121a2 has a flare shape which spreads downward when viewed from the side. The conductive ring 123 is supported by the insulating ring 121, and has at least three grooves 123a corresponding to at least three through holes 121a, respectively on the bottom thereof. Each of at least three grooves 123a has the third width W3 in the radial direction R2 of the conductive ring 123 and the fourth width W4 in the circumferential direction C2 of the conductive ring 123. The fourth width W4 is smaller than the third width W3. At least three lift pins 131 are placed below the ring supporting surface 110b, and correspond to at least three grooves 123a, respectively. Each of at least three lift pins 131 has the upper supporting portion 131a and the lower supporting portion 131b. The upper supporting portion 131a is configured to support the conductive ring 123 from downward by contacting the bottom 123a1 of the groove 123a of the conductive ring 123 via the through hole 121a of the insulating ring 121. The lower supporting portion 131b is configured to support the insulating ring 121 from downward by contacting the inclined surface 1211a of the insulating ring 121 defining the lower hole portion 121a2. At least one actuator 132 is configured to move at least three lift pins 131 in the vertical direction. The lower supporting portion 131b of the lift pin 131 is configured to 2-point-contact the inclined surface 1211a in the radial direction R1 of the insulating ring 121 without contacting the inclined surface 1211a in the circumferential direction C1 of the insulating ring 121.

In an exemplary embodiment, each of at least three through holes 121a of the insulating ring 121 has the rectangular shape when viewed from upward. In an exemplary embodiment, the upper supporting portion 131a has a cylindrical shape having a first diameter W5 smaller than the second width W2, and the lower supporting portion 131b has a cylindrical shape having a second diameter W6 larger than the second width W2. In an exemplary embodiment, the tip of the upper supporting portion 131a of the lift pin 131 has the hemispherical shape 131a1. In an exemplary embodiment, the tip of the upper supporting portion 131a of the lift pin 131 has the flat surface 131a2. In an exemplary embodiment, each of at least three grooves 123a of the conductive ring 123 is defined by the flat bottom 123a1', the vertical side surface 123a3 which extends downward from the flat bottom 123a1', and the flare-shaped side surface 123a2 which spreads downward from the vertical side surface 123a3. In an exemplary embodiment, each of at least three grooves 123a of the conductive ring 123 is defined by the curved bottom 123a1'. In an exemplary embodiment, each of at least three grooves 123a of the conductive ring 123 is defined by the flat bottom 123a1' and the flare-shaped side surface 123a2 which spreads downward from the flat bottom 123a1'. In an exemplary embodiment, the plasma processing system 1 further includes an additional conductive ring 122 placed to surround around the conductive ring 123 and the insulating ring 121. Further, at least a part of the conductive ring 122 overlaps with an external portion of the insulating ring 121 in the vertical direction.

Further, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber 10, a substrate support 11, a first annular member 121, a second annular member 123, at least three lift pins 131, and at least one actuator 132. The substrate support 11 is placed in the plasma processing chamber 10, and has a substrate supporting surface 110a and a ring supporting surface 110b. The first annular member 121 is placed on the ring supporting surface 110b, and has at least three through holes 121a. Each of at least three through holes 121a has an

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upper hole portion **121a1** and a lower hole portion **121a2**. The upper hole portion **121a1** has a first width **W1** in a first direction **R1** and a second width **W2** in a second direction **C1** orthogonal to the first direction **R1**. The second width **W2** is smaller than the first width **W1**. The lower hole portion **121a2** has the flare shape **1211a** which spreads downward when viewed from the side. The second annular member **123** is supported by the first annular member **121**, and has at least three grooves **123a** corresponding to at least three through holes **121a**, respectively on the bottom thereof. Each of at least three grooves **123a** has a third width **W3** in a third direction **R2** and a fourth width **W4** in a fourth direction **C2** orthogonal to the third direction **R2**. The fourth width **W4** is smaller than the third width **W3**. At least three lift pins **131** are placed below the ring supporting surface **110b**, and correspond to at least three grooves **123a**, respectively. Each of at least three lift pins **131** has the upper supporting portion **131a** and the lower supporting portion **131b**. The upper supporting portion **131a** is configured to support the second annular member **123** from downward by contacting the bottom **123a1** of the groove **123a** of the second annular member **123** via the through hole **121a** of the first annular member **121**. The lower supporting portion **131b** is configured to support the first annular member **121** from downward by contacting the inclined surface **1211a** of the first annular member **121** defining the lower hole portion **121a2**. At least one actuator **132** is configured to move at least three lift pins **131** in the vertical direction.

In the plasma processing system, the substrate **W** is loaded into the plasma processing apparatus **1** by using the transfer apparatus **2**, and desired plasma processing such as etching is performed for the loaded substrate **W**, and then the substrate **W** is unloaded to the outside of the plasma processing apparatus **1** by using the transfer apparatus **2**.

In the exemplary embodiment, the inner edge ring **123** and the outer edge ring are made of a conductive material. Therefore, at the time of the plasma processing in the plasma processing system, a sheath is formed above the ring assembly **120**, more specifically, above the inner edge ring **123** as the conductive ring and the outer edge ring **122** as the additional conductive ring. Therefore, an excellent and uniform plasma processing result may be obtained at the center and the periphery of the substrate **W** held on the substrate support **11**.

Here, as described above, in recent years, the plasma processing system, enhancement of installation precision of the ring assembly on the substrate support **11** is required.

In the Japanese Laid-open Patent Publication, it is disclosed that in order to enhance the installation precision of the annular member, the groove formed in the edge ring and the through hole formed in the cover ring are formed to be larger than the transfer precision of the annular member, but in this case, for example, there is a risk that the positioning precision of the annular member will deteriorate when a relative dislocation occurs at positions of the lifter and the groove (through hole) due to a mechanical difference (e.g., processing tolerance) or thermal expansion of the annular member.

In this point, by the plasma processing apparatus **1** according to the technique of the present disclosure, the through hole **121a** is formed in the insulating ring **121** corresponding to the cover ring in a substantially rectangular shape in which the first width **W1** is larger than the second width **W2**, and the groove **123a** is also formed in the inner edge ring **123** in a substantially long hole shape in which the third width **W3** is larger than the fourth width **W4**.

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Therefore, even though the position of the lift pin **131** and the position of the through hole **121a** (groove **123a**) are dislocated due to the processing tolerance, the insulating ring **121** is slidably movable with the lift pin **131** as an axis, i.e., becomes a free state in the long-side direction of the through-hole **121a** (groove **123a**), so the insulating ring **121** may be appropriately guided to a desired location. Further, in this case, since the outer edge ring **122** as the additional conductive ring is held on the upper surface of the insulating ring **121**, the location of the outer edge ring **122** is simultaneously corrected.

Further, in this case, as described above, the lower supporting portion **131b** of the lift pin **131** contacts two points on the inclined portion of the lower hole portion **121a2** of the through hole **121a**. As a result, as compared with a case where the lower supporting portion **131b** and the inclined portion of the lower hole portion **121a2** are in contact with each other at three points or more, or in line-contact with each other, and in surface-contact with each other, the insulating ring **121** easily slidably moves to more appropriately execute dislocation correction of the insulating ring **121** (ring assembly **120**).

Further, in this case, as described above, the upper supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** of the groove **123a**.

When the upper supporting portion **131a** and the bottom **123a1** of the groove **123a** are in contact with each other only at one point or two points, since the upper supporting portion **131a** and the bottom **123a1** slidably move relatively easily, the dislocation correction of the inner edge ring **123** may be facilitated, but meanwhile, since the upper supporting portion **131a** and the bottom **123a1** slidably move easily, the dislocation of the inner edge ring **123** easily occurs when the lift pin **131** moves in the vertical direction.

Meanwhile, when the upper supporting portion **131a** and the bottom **123a1** of the groove **123a** are in surface-contact with each other, relative slidable movement of the bottom **123a1** and the upper supporting portion **131a** may be suppressed, but meanwhile, since the slidable movement is suppressed, it is difficult to correct the dislocation of the inner edge ring **123**.

In this point, in the above exemplary embodiment, since the upper supporting portion **131a** of the lift pin **131** is in line-contact with the bottom **123a1** of the groove **123a**, the bottom **123a1** slidably moves on the upper supporting portion **131a** to appropriately execute the dislocation correction of the inner edge ring **123** and at the same time, the dislocation of the inner edge ring **123** may also be appropriately suppressed when the lift pin **131** moves in the vertical direction.

Further, when the through hole **121a** is formed in the insulating ring **121**, the through hole having the substantially rectangular shape in which the first width **W1** is larger than the second width **W2** as such, the insulating ring **121** slidably moves in the long-side direction of the through hole **121a** as described above, and as a result, there is a risk that the location of the insulating ring **121** will be dislocated when the lift pin **131** moves in the vertical direction.

By considering such a point, the through hole **121a** is preferably formed in the substantially square shape as described above in order to suppress the dislocation of the insulating ring **121** (enhance the placement precision of the insulating ring **121**). Further, in this case, the long side-direction width (second width **W2**) of the through hole **121a** is preferably set to a size in which the lower supporting portion **131b** of the lift pin **131** contacts the inclined portion of the through hole **121a** at two points.

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As a result, the lower supporting portion 131b and the inclined portion contact at least two points each other to execute the dislocation correction of the insulating ring 121, and at the same time, a free width of the lift pin 131 in the long-side direction of the through hole 121a decreases to suppress the dislocation of the insulating ring 121 when the lift pin 131 moves in the vertical direction.

<Replacement Method of Ring Assembly>

Next, replacement processing of the ring assembly 120 configured as above will be described by using drawings. Further, the following processing is executed under the control by the controller 3 as an example.

(Step St1: Delivery and Unloading of Inner Edge Ring 123)

First, a consumed inner edge ring 123 to be replaced is delivered to the transfer apparatus 2 from the substrate support 11.

Specifically, first, ascending of the lifter 130 is executed, and as illustrated in FIG. 7, the inner edge ring 123 is delivered to the upper supporting portion 131a of the lifter 130 passing through the through hole 113a of the support 113 and the through hole 121a of the insulating ring 121 from the upper surface of the insulating ring 121. In this case, the ascending of the lifter 130 is executed in a range in which the insulating ring 121 is not delivered by the lower supporting portion 131b, and further, executed until the tip of the upper supporting portion 131a reaches a desired height H1. Here, the desired height H1 is a height of the upper supporting portion 131a of the lifter 130 based on the upper surface of the outer edge ring 122 held on the insulating ring 121 mounted on the support 113. Further, the desired height H1 is a height at which the transfer arm 2a does not interfere with the outer edge ring 122 and the inner edge ring 123 when the transfer arm 2a is inserted and extracted between the outer edge ring 122 on the insulating ring 121 mounted on the support 113 and the inner edge ring 123 supported on the upper supporting portion 131a.

Next, the inner edge ring 123 is unloaded from the plasma processing chamber 10 of the plasma processing apparatus 1.

Specifically, the transfer arm 2a is inserted into the decompressed plasma processing chamber 10 via a loading/unloading port (not illustrated). In addition, the transfer arm 2a moves between the outer edge ring 122 on the insulating ring 121 mounted on the support 113 and the inner edge ring 123 supported on the upper supporting portion 131a of the lifter 130.

Subsequently, descending of the lifter 130 is executed, and the inner edge ring 123 is delivered to the transfer arm 2a from the upper supporting portion 131a of the lifter 130. Next, the transfer arm 2a is extracted from the plasma processing chamber 10, and the inner edge ring 123 itself is unloaded to the outside of the plasma processing apparatus 1. The unloaded inner edge ring 123 is loaded into a receiving module (not illustrated) as an example.

(Step St2: Delivery and Unloading of Insulating Ring 121 and Outer Edge Ring 122)

When the inner edge ring 123 is unloaded from the plasma processing apparatus 1, the consumed insulating ring 121 and outer edge ring 122 to be replaced are subsequently delivered to the transfer apparatus 2 from the substrate support 11.

Specifically, first, the ascending of the lifter 130 is executed, and as illustrated in FIG. 8, the insulating ring 121 is delivered to the lower supporting portion 131b of the lifter 130 passing through the through hole 113a of the support 113 from the upper surface of the annular region 110b (ring supporting surface) of the substrate support 11. In this case,

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since the outer edge ring 122 is held on the insulating ring 121, the outer edge ring 122 is also delivered by the lower supporting portion 131b via the insulating ring 121. In this case, the ascending of the lifter 130 is executed until the step of the lower supporting portion 131b reaches a desired height H2. Here, the desired height H2 is a height of the lower supporting portion 131b of the lifter 130 based on the upper surface (more specifically, the upper surface of the electrostatic chuck 112) of the substrate support 11. Further, the desired height H2 is a height at which the transfer arm 2a does not interfere with the insulating ring 121 and the electrostatic chuck 112 when the transfer arm 2a is inserted and extracted between the electrostatic chuck 112 and the insulating ring 121 supported on the lower supporting portion 131b.

Next, the insulating ring 121 and the outer edge ring 122 are unloaded from the plasma processing chamber 10 of the plasma processing apparatus 1.

Specifically, the transfer arm 2a is inserted into the decompressed plasma processing chamber 10 via a loading/unloading port (not illustrated). In addition, the transfer arm 2a moves between the electrostatic chuck 112, and the insulating ring 121 supported on the lower supporting portion 131b of the lifter 130.

Subsequently, the descending of the lifter 130 is executed, and the insulating ring 121 and the outer ring 122 are delivered to the transfer arm 2a from the lower supporting portion 131b of the lifter 130. Next, the transfer arm 2a is extracted from the plasma processing chamber 10, and the insulating ring 121 and the outer edge ring 122 are unloaded to the outside of the plasma processing apparatus 1. The unloaded insulating ring 121 and the outer edge ring 122 are loaded into a receiving module (not illustrated) as an example.

(Step St3: Loading and Mounting of Ring Assembly 120)

When the inner edge ring 123, the insulating ring 121, and the outer edge ring 122 are unloaded from the plasma processing apparatus 1, a new ring assembly 120 (the insulating ring 121, the outer edge ring 122, and the inner edge ring 123) for replacement are subsequently loaded into the plasma processing apparatus 1.

Specifically, the transfer arm 2a of the transfer apparatus 2, which holds the new ring assembly 120 including the insulating ring 121, and the outer edge ring 122 and the inner edge ring 123 held on the insulating ring 121 is inserted into the plasma processing chamber 10 via the loading/unloading port (not illustrated). In this case, the plasma processing chamber 10 may be decompressed. In addition, as illustrated in FIG. 9, the new ring assembly 120 is transferred by the transfer arm 2a above the annular region 110b of the substrate support 11.

Next, the new ring assembly 120 is mounted on the annular region 110b of the substrate support 11 from the transfer apparatus 2.

Specifically, the ascending of the lifter 130 is executed, and as illustrated in FIG. 10A, the inner edge ring 123 is delivered to the upper supporting portion 131a of the lifter 130 passing through the through hole 121a of the insulating ring 121 from the upper surface of the insulating ring 121 held on the transfer arm 2a.

In this case, in the through hole 121a of the insulating ring 121 and the groove 123a of the inner edge ring 123, as described above, the lower end of the inclined portion is formed to be at least larger than the transfer precision of the inner edge ring 123 by the transfer apparatus 2. As a result, even though the lift pin 131, the through hole 121a, and the groove 123a are dislocated due to the transfer precision of

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the transfer apparatus 2, the lift pin 131 is appropriately inserted into the through hole 121a to be further guided to the bottom 123a1 of the groove 123a.

Subsequently, the ascending of the lifter 130 is continued, and the insulating ring 121 is delivered to the lower supporting portion 131b of the lifter 130 from the transfer arm 2a as illustrated in FIG. 10B. In this case, the outer edge ring 122 on the insulating ring 121 is also simultaneously delivered to the lower supporting portion 131b. In this case, the ascending of the lifter 130 is executed until the step of the lower supporting portion 131b reaches a desired height H2.

In this case, in the exemplary embodiment, in the through hole 121a of the insulating ring 121 and the groove 123a of the inner edge ring 123, a diameter-direction width (a first width W1 or a third width W3) is larger a circumferential width (a second width W2 or a fourth width W4). As a result, even though the lift pin 131, the through hole 121a, and the groove 123a are dislocated due to the processing tolerance, the lift pin 131 is appropriately inserted into the through hole 121a to be further guided to the bottom 123a1 of the groove 123a.

Further, in this case, the lower supporting portion 131b of the lift pin 131 contacts the inclined portion of the through hole 121a at two points, and as a result, the insulating ring 121 becomes a free state in the long-side direction of the through hole 121a with the lift pin 131 as the axis. As a result, even though the lift pin 131, the through hole 121a, and the groove 123a are dislocated due to various causes, the insulating ring 121 slidably moves in the long-side direction of the through hole 121a to appropriately position the insulating ring 121 and the outer edge ring 122 at desired locations.

Further, similarly, the upper supporting portion 131a of the lift pin 131 is in line-contact with the bottom 123a1 of the groove 123a, and as a result, the inner edge ring 123 may be appropriately positioned at a desired location, and at the same time, the dislocation of the inner edge ring 123 at the time of subsequent mounting of the ring assembly 120 may be suppressed.

Next, extraction of the transfer arm 2a from the plasma processing chamber 10 is executed. Further, the descending of the lifter 130 is executed. As a result, the insulating ring 121, the outer edge ring 122, and the inner edge ring 123 are mounted on the upper surface of the annular region 110b of the substrate support 11 and the upper surface of the support 113. Specifically, first, the insulating ring 121 and the outer edge ring 122 are mounted on the upper surface of the support 113, and next, the inner edge ring 123 is mounted on the upper surface of the insulating ring 121 and the upper surface of the annular region 110b of the substrate support 11.

Therefore, a series of processing for mounting the ring assembly 120 on the substrate support 11 is completed.

Further, in the exemplary embodiment, loading and mounting of the insulating ring 121, the outer edge ring 122, and the inner edge ring 123 as the ring assembly 120 are simultaneously executed, but loading and mounting of any one of the insulating ring 121, the outer edge ring 122, and the inner edge ring 123 may be independently executed.

Further, in the exemplary embodiment, delivery and unloading of the insulating ring 121, the outer edge ring 122, and the inner edge ring 123 as the ring assembly 120 are sequentially executed, but delivery and unloading of any one of the insulating ring 121, the outer edge ring 122, and the inner edge ring 123 may be simultaneously executed. In this case, in a state in which the inner edge ring 123 is mounted on the insulating ring 121, i.e., the delivery between the lift

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pin 131 and the transfer arm 2a and unloading from the plasma processing apparatus 1 may be executed integrally with the insulating ring 121.

<Modified Example of Edge Ring>

FIGS. 11A to 11C are diagrams illustrating a configuration example of an inner edge ring 223 according to another exemplary embodiment.

In the inner edge ring 223 according to another exemplary embodiment, among a plurality of, in the illustrated example, three grooves 223a to 223c formed on the bottom, at least one groove (a groove 223a in the illustrated example) may be formed in a circular shape (FIG. 11B) and the remaining grooves (grooves 223b and 223c in the illustrated example) may be formed in a long hole shape (FIG. 11C).

In an exemplary embodiment, the groove 223a having the circular shape has the substantially same diameter (a first diameter D1 of FIGS. 11A and 11B) as the width (fifth width W5) of the upper supporting portion 131a of the lift pin 131 on a bottom 223a1. Further, the bottom 223a1 of the groove 223a has a shape of being fitted into a tip shape of the upper supporting portion 131a of the lift pin 131. In other words, the tip of the upper supporting portion 131a is configured to be in surface-contact with the groove 223a1 of the groove 223a formed in the inner edge ring 223.

Further, a side surface 223a2 of the groove 223a has an inclined portion is so that the first diameter D1 of the bottom 223a1 spreads downward (toward the bottom of the inner edge ring 223). A lower end of the inclined portion of the groove 223a is preferably formed to be at least larger than the transfer precision of the inner edge ring 223 by the transfer apparatus 2.

In an exemplary embodiment, the grooves 223b and 223c having the long hole shape have the same structure as the groove 123a of the inner edge ring 123 illustrated as the exemplary embodiment. That is, the grooves 223b and 223c are formed in which a third width W3 which extends in the radial direction of the inner edge ring 223 is larger than a fourth width W4 which extends in the circumferential direction of the inner edge ring 223 orthogonal to the radial direction, and bottoms 223b1 and 223c1 have, in particular, a shape of being fitted into the tip shape of the upper supporting portion 131a of the lift pin 131 in an extension direction of the fourth width W4. Further, the inclined portions are formed on side surfaces 223b2 and 223c2 of the grooves 223b and 223c. A lower end of the inclined portion of the groove 123a is preferably formed to be at least larger than the transfer precision of the inner edge ring 123 by the transfer apparatus 2.

As illustrated as the exemplary embodiment, when all of three grooves 123a formed on the bottom of the inner edge ring 123 have the long hole shape, the inner edge ring 123 becomes the free state in the long-side direction of the groove 123a as described above, so there is a risk that the inner edge ring 123 slidably moves to be dislocated.

Therefore, in the inner edge ring 223 according to another exemplary embodiment, at least one groove 223a is formed in the circular shape, and positioning of the inner edge ring 223 is executed based on the circular groove 223a. That is, the plasma processing system according to an exemplary embodiment includes a plasma processing chamber 10, a substrate support 11, an annular member 223, at least three lift pins 131, and at least one actuator 132. The substrate support 11 is placed in the plasma processing chamber 10, and has a substrate supporting surface 110a and a ring supporting surface 110b. The annular member 223 is placed on the ring supporting surface 110b, and has at least three

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grooves **223a** to **223c**. A first groove **223a** among at least three grooves **223a** to **223c** has the circular bottom **223a** when viewed from upward. A second groove **223b** and a third groove **223c** among at least three grooves **223a** to **223c** have the bottoms **223b1** and **223c1** having a first width **W3** in a radial direction **R3** of the annular member **223** and a second width **W4** in a circumferential direction **C3** of the annular member **223**. The second width **W4** is smaller than the first width **W3**. At least three cylindrical lift pins **131** are placed below the ring supporting surface **110b**, and correspond to at least three grooves **223a** to **223c**, respectively. At least three cylindrical lift pins **131** are configured to support the annular member **223** from downward by contacting the bottoms of the grooves **223a** to **223c** of the annular member **223**, respectively. At least one actuator **132** is configured to move at least three cylindrical lift pins **131** in the vertical direction.

Therefore, the tip of the upper supporting portion **131a** and the bottom **223a1** of the groove **223a** having the substantially same size are fitted to be in surface-contact with each other, so the inner edge ring **223** and the lifter **130** are at least suppressed from slidably moving in a horizontal direction. Further, since the remaining grooves **223b** and **223c** are formed in the long hole shape, even though the grooves **223b** and **223c**, and the lift pin **131** are dislocated due to various causes, the lift pin **131** may be appropriately guided to the grooves **223b** and **223c** by correcting the dislocation and the inner edge ring **223** may be positioned at a desired location.

Further, in the exemplary embodiment, among at least three grooves **223a** to **223c**, one groove **223a** is formed in the circular shape, and the remaining grooves **223b** and **223c** are formed in the long hole shape, but for example, when the positioning of the inner edge ring **223** may be appropriately executed, each of at least three grooves **223a** to **223c** may be formed in the circular shape.

In other words, in an exemplary embodiment, each of at least three grooves **223a** to **223c** of the conductive ring **123** is defined by the bottom **223a1** having a shape to be fitted into the tip of the upper supporting portion **131a** of the lift pin **131** and the flare-shaped side surface **223a2** which spreads downward from downward surface.

<Modified Example of Groove Formed in Inner Edge Ring>

Further, sectional shapes of long hole shaped grooves (the groove **123a** of the inner edge ring **123** and the grooves **223b** and **223c** of the inner edge ring **223**) formed in the inner edge ring are not limited to the exemplary embodiment.

For example, as illustrated in FIG. 12A, the groove **123a** may have the bottom **123a1**, the vertical side surface **123a3** as a vertical wall which extends vertically downward from the flat bottom **123a1**, and the flare-shaped side surface **123a2** as extended wall which spreads downward from the vertical side surface **123a3**. In this case, a short side-direction fourth width **W4** (a width of the vertical wide surface **123a3** part) of the groove **123a** is preferably formed to be a little larger than the diameter (the fifth width **W5**) of the upper supporting portion **131a** to slidably move with respect to the lift pin **131**. In this case, a contact scheme of the upper supporting portion **131a** and the groove **123a** is a 1-point contact.

For example, as illustrated in FIG. 12B, the groove **123a** may be formed on a curved surface having a larger curvature than the tip (hemispherical shape **131a1**) of the hemispherical lift pin **131**. In other words, the groove **123a** has only the curved bottom **123a1** having the curved shape, and may not include the vertical side surface **123a3** and the flare-shaped side surface **123a2**. In this case, the groove **123a** is config-

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ured to slidably move with respect to the lift pin **131**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is the 1-point contact.

For example, as illustrated in FIG. 12C, the groove **123a** may have the bottom **123a1** having the substantially flat surface and the flare-shaped side surface **123a2** as the extended wall which spreads downward from the flat bottom **123a1**. In other words, as compared with the exemplary embodiment, in the example illustrated in FIG. 12C, the flat bottom **123a1** is not formed in a shape of being fitted into the lift pin **131**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is a 3-point contact.

Further, in the examples illustrated in FIGS. 5 and FIGS. 12A to 12C above, the flare-shaped side surface **123a2** which spread downward from downward surface **123a1** (or vertical side surface **123a3**) of the groove **123a** is configured by a side surface having a liner shape when viewed from the side, which spreads at a predetermined ratio downward, but the flare-shaped side surface **123a2** may be configured by the side surface having the curved shape when viewed from the side.

Specifically, as illustrated in FIG. 12D as an example, the groove **123a** may have a curved bottom **123a1** having the curved shape, which has a larger curvature than the tip (hemispherical shape **131a1**) of the hemispherical lift pin **131**, and a flare-shaped curved side surface **123a4** which spreads downward from the curved bottom **123a1** and which is curved upward (toward an opposite side to the lift pin **131**). In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is the 1-point contact.

Further, the curvature of the flare-shaped curved side surface **123a4** is not particularly limited. In other words, the groove **123a** is formed in the inner edge ring **123** provided in the plasma processing apparatus **1** according to the technique of the present disclosure, and the groove **123a** is constituted by the bottom **123a1**, and one or more side surfaces (side surfaces **123a2**, **123a3**, and **123a4**) having a predetermined curvature of "0" or more. The curvature of side surface being "0" refers to the side surface being the linear shape (side surface **123a2**) when viewed from the side. The curvature being larger than "0" refers to the side surface being the curved shape (side surface **123a4**) when viewed from the side.

As such, the groove **123a** (grooves **223b** and **223c**) may be formed in a predetermined sectional shape. In this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** varies depending on the sectional shape. From a viewpoint of facilitating the positioning of the upper supporting portion **131a** and the groove **123a**, a contact area of the upper supporting portion **131a** and the groove **123a** is preferably small. From a viewpoint of enhancing the installation precision of the upper supporting portion **131a** and the groove **123a**, the contact area of the upper supporting portion **131a** and the groove **123a** is preferably large.

As such, the sectional shape of the groove **123a** (grooves **223b** and **223c**) may be appropriately designed according to a purpose.

Further, the sectional shape of the circular groove (the groove **223a** of the inner edge ring **223**) formed in the inner edge ring is not also limited to the exemplary embodiment.

For example, as illustrated in FIG. 13A, the groove **223a** may have a bottom **223a1**, a vertical side surface **223a3** as a vertical wall which extends vertically downward from the flat bottom **223a1**, and a flare-shaped side surface **223a2** as extended wall which spreads downward from the vertical

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side surface **223a3**. In this case, a first diameter **D1** (a width of the vertical side surface (**223a3**) part) of the groove **223a** is preferably formed to be a little larger than the diameter (fifth width **W5**) of the upper supporting portion **131a** in order to appropriately contact the tip of the lift pin **131** with the flat bottom **223a1'**. Further, in this case, the contact scheme of the upper supporting portion **131a** and the groove **123a** is the 1-point contact.

For example, as illustrated in FIG. 13B, the groove **223a** may be formed on a curved surface having a larger curvature than the tip of the hemispherical lift pin **131**. In other words, the groove **223a** has only the curved bottom **223a1'** having the curved shape, and may not include the flare-shaped side surface **223a2** and the vertical side surface **223a3**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **223a** is the 1-point contact.

For example, as illustrated in FIG. 13C, the groove **223a** may have the bottom **223a1''** having the substantially flat surface and the flare-shaped side surface **223a2** as the extended wall which spreads downward from the flat bottom **223a1''**. In other words, as compared with the example illustrated in FIG. 11, in the example illustrated in FIG. 13C, the flat bottom **223a1''** is not formed in a shape of being fitted into the lift pin **131**. In this case, the contact scheme of the upper supporting portion **131a** and the groove **223a** is the 3-point contact.

Further, the illustrated is omitted, but even for the circular groove **223a**, similarly to the example illustrated in FIG. 12D, the flare-shaped side surface **123a2** may be configured by the curve-shaped side surface when viewed from the side. In other words, the circular groove **223a** may have a flare-shaped curved side surface not illustrated.

Hereinabove, various exemplary embodiments have been described, but the present disclosure is not limited to the exemplary embodiment, but various additions, omissions, substitutions, and modifications may be made. Further, it is possible to form another exemplary embodiment by combining elements in different exemplary embodiments.

The invention claimed is:

1. A plasma processing apparatus, comprising:

a plasma processing chamber;

a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface;

an insulating ring disposed on the ring supporting surface, the insulating ring having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a radial direction of the insulating ring and a second width in a circumferential direction of the insulating ring, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side;

a conductive ring supported by the insulating ring, the conductive ring having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a radial direction of the conductive ring and a fourth width in a circumferential direction of the conductive ring, the fourth width being smaller than the third width;

at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting por-

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tion being configured to support the conductive ring from downward by contacting a bottom of the groove of the conductive ring through the through hole of the insulating ring, the lower supporting portion being configured to support the insulating ring from downward by contacting an inclined surface of the insulating ring, the inclined surface defining the lower hole portion; and

at least one actuator configured to vertically move the at least three lift pins.

2. The plasma processing apparatus of claim 1, wherein the lower supporting portion of the lift pin is configured to come into contact with two points on the inclined surface in the radial direction of the insulating ring without contacting the inclined surface in the circumferential direction of the insulating ring.

3. The plasma processing apparatus of claim 1, wherein each of the at least three through holes of the insulating ring has a rectangular shape when viewed from upward.

4. The plasma processing apparatus of claim 1, wherein the upper supporting portion has a cylindrical shape having a first diameter smaller than the second width, and the lower supporting portion has a cylindrical shape having a second diameter larger than the second width.

5. The plasma processing apparatus of claim 1, wherein a tip of the upper supporting portion of the lift pin has a hemispherical shape.

6. The plasma processing apparatus of claim 1, wherein a tip of the upper supporting portion of the lift pin has a flat surface.

7. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by:

a flat bottom,

a vertical side surface which extends downward from the flat bottom, and

a flare-shaped side surface which spreads downward from the vertical side surface.

8. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by a curved bottom.

9. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring is defined by:

a flat bottom, and

a flare-shaped side surface which spreads downward from the flat bottom.

10. The plasma processing apparatus of claim 1, wherein each of the at least three grooves of the conductive ring has a bottom having a shape to be fitted into a tip of the upper supporting portion of the lift pin, and a flare-shaped side surface which spreads downward from the bottom.

11. The plasma processing apparatus of claim 1, further comprising:

an additional conductive ring disposed to surround a circumference of the conductive ring and the insulating ring,

wherein at least a part of the additional conductive ring overlaps an outer portion of the insulating ring in a vertical direction.

12. A plasma processing apparatus, comprising:

a plasma processing chamber;

a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface;

an annular member disposed on the ring supporting surface, the annular member having at least three

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grooves on a lower surface, a first groove among the at least three grooves having a circular bottom when viewed from upward, a second groove and a third groove among the at least three grooves having a bottom having a first width in a radial direction of the annular member and a second width in a circumferential direction of the annular member, the second width being smaller than the first width;

at least three cylindrical lift pins disposed below the ring supporting surface, the at least three cylindrical lift pins corresponding to the at least three grooves, respectively, each of the at least three cylindrical lift pins being configured to support the annular member from downward by contacting the bottom of the groove of the annular member; and

at least one actuator configured to vertically move the at least three cylindrical lift pins.

13. The plasma processing apparatus of claim 12, wherein a tip of the cylindrical lift pin has a hemispherical shape.

14. The plasma processing apparatus of claim 12, wherein a tip of the cylindrical lift pin has a flat surface.

15. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by:

- a flat bottom,
- a vertical side surface which extends downward from the flat bottom, and
- a flare-shaped side surface which spreads downward from the vertical side surface.

16. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by a curved bottom.

17. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular member is defined by:

- a flat bottom, and
- a flare-shaped side surface which spreads downward from the flat bottom.

18. The plasma processing apparatus of claim 12, wherein each of the at least three grooves of the annular conductive ring has a bottom having a shape to be fitted into a tip of a

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upper supporting portion of the cylindrical lift pin, and a flare-shaped side surface which spreads downward from the bottom.

19. A plasma processing apparatus, comprising:

- a plasma processing chamber;
- a substrate support disposed in the plasma processing chamber, the substrate support having a substrate supporting surface and a ring supporting surface;
- a first annular member disposed on the ring supporting surface, the first annular member having at least three through holes, each of the at least three through holes having an upper hole portion and a lower hole portion, the upper hole portion having a first width in a first direction and a second width in a second direction orthogonal to the first direction, the second width being smaller than the first width, the lower hole portion having a flaring shape spreading downward when viewed from a side;
- a second annular member supported by the first annular member, the second annular member having at least three grooves on a lower surface, the at least three grooves corresponding to the at least three through holes, respectively, each of the at least three grooves having a third width in a third direction and a fourth width in a fourth direction orthogonal to the third direction, the fourth width being smaller than the third width;
- at least three lift pins disposed below the ring supporting surface, the at least three lift pins corresponding to the at least three grooves, respectively, each of the at least three lift pins having an upper supporting portion and a lower supporting portion, the upper supporting portion being configured to support the second annular member from downward by contacting a bottom of the groove of the second annular member through the through hole of the first annular member, the lower supporting portion being configured to support the first annular member from downward by contacting an inclined surface of the first annular member, the inclined surface defining the lower hole portion; and
- at least one actuator configured to vertically move the at least three lift pins.

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